

INVITED REVIEW

Timing of enteral nutrition and parenteral nutrition in the PICU

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Abstract

The timing of nutrition support initiation has the potential to positively impact nutrition and clinical outcomes in infants and children with critical illness. Early enteral nutrition within 24–48 h and attainment of both a 60% energy and protein goal by the end of the first week of pediatric intensive care unit admission are reported to be significantly associated with improved survival in large observational studies. The results of one randomized controlled trial demonstrated increased morbidity in infants and children with critical illness assigned to early vs delayed supplemental parenteral nutrition. Observational studies in this population also suggest increased mortality with exclusive parenteral nutrition and worse nutrition outcomes when parenteral nutrition is delayed. Subsequently, current nutrition support guidelines recommend early enteral nutrition and avoidance of early parenteral nutrition, although the available evidence used to create the guidelines was inadequate to inform bedside nutrition support practice to improve outcomes. These guidelines are limited by the included studies with small numbers and heterogeneity of patients and research design that confound study outcomes and interpretation. This article provides a narrative review of the timing of nutrition support on outcomes in infants and children with critical illness, strategies to optimize timing and adequacy of nutrition support, and literature gaps, including the timing of parenteral nutrition initiation for children with malnutrition and those with contraindications to enteral nutrition and accurate measurement of energy requirements.

KEYWORDS

Children, critical illness, enteral nutrition, newborn infants, parenteral nutrition, pediatric intensive care units

INTRODUCTION

The time from pediatric intensive care unit (PICU) admission to the initiation of nutrition support and achievement of the target energy and protein intake has

the potential to impact the outcomes of children with critical illness. The Society of Critical Care Medicine and the American Society for Parenteral and Enteral Nutrition (SCCM-ASPEN) jointly advocate for the initiation of enteral nutrition (EN) within 24–48 h from admission to

the PICU in infants and children from 1 month of age through 17 years with critical illness.¹ These guidelines also recommend avoidance of parenteral nutrition (PN) within the first 24 h of admission. European Society of Pediatric and Neonatal Intensive Care guidelines for term infants older than 37 weeks' gestation and children up to 18 years of age recommend EN within 24 h of admission while withholding PN during the first week and providing micronutrients.² Published nutrition support guidelines for children with critical illness exclude patients with burn injuries and are limited by the dearth of literature, small sample sizes, and heterogeneity of patient population, illness severity, study design, and nutrition status that confound the outcomes and interpretation of the studies used to inform bedside practice. This narrative review provides an updated evaluation focusing on the impact of the timing of nutrition support interventions, specifically EN and PN, on outcomes in term infants and children from 1 month to 17 years of age with critical illness. Strategies to improve gastrointestinal tolerance to EN, avoid common barriers to the adequate administration of nutrition support, and increase energy and protein intake to meet targets and potentially optimize outcomes are also discussed. In this article, nutrition support refers to the provision of EN and PN, with EN delivered as a formula or human milk into the gastrointestinal tract by an enteral access device, and PN with macronutrients, micronutrients, fluid, and electrolytes administered via central or peripheral venous catheter.

SEARCH STRATEGY

Literature published from January 1991 through November 2022 was screened from MEDLINE/PubMed, EMBASE, and Google Scholar for citations relevant to the timing of nutrition support in pediatric patients from 1 month through 17 years of age with critical illness. Citations included in this review were randomized controlled trials, meta-analyses, observational studies, and systematic reviews, and nutrition support guidelines for children with critical illness, narrative reviews, surveys, and commentaries were used for the introduction section and discussion of the studies in this article. Search terms included "critical illness," "intensive care," "infant," "child," "adolescent," "pediatric," and were combined (and, and/or, or) with terms such as "nutrition support," "enteral nutrition," "parenteral nutrition," "energy," "macronutrient," "protein," "carbohydrate," "fat," "strategies," "barriers," "timing," "initiation," "target," "goal," "early," "late," "immediate," "delayed," "adequate," "inadequate," "optimal,"

"suboptimal," "underfeeding," "overfeeding," "exclusive," "supplemental," and "outcome." Adults, preterm infants with corrected gestational age younger than 1 month, and term neonates younger than 1 month of age were excluded. Appropriate delivery of fluid, electrolytes, and micronutrients, including vitamins and trace elements, are essential for the optimal nutrition care of children with critical illness but are not examined in this review.

TIMING OF EN INITIATION

EN is the preferred method of nutrition support in children with critical illness.¹ Observational studies that demonstrate outcome improvements associated with the timing and amount of EN form the basis of the SCCM-ASPEN guidelines.¹ Limited evidence is available to determine the benefits of early vs delayed EN, as shown in Table 1; furthermore, the definition of early EN differs across studies, making comparisons difficult. The effect of early EN in 6–24 h vs EN delayed after 24 h on length of PICU stay was recently examined in an open-label single-center randomized controlled trial.³ Sixty children assigned to the early group initiated EN in a median of 6 h, whereas the 60 randomly assigned to the late group commenced in a median of 40 h. No significant difference in PICU length of stay or other clinical outcomes was demonstrated with early vs delayed EN; however, the study lacked adequate power to detect differences between groups. Outcomes were also confounded by the baseline differences in group demographics, as the early EN patients were younger with a higher proportion of malnutrition compared with the late EN group. Further research is needed in this population to determine the optimal timing of EN to benefit outcomes.

Additional randomized controlled trials of early vs delayed EN in children with critical illness are limited to children with burn injuries. These investigations demonstrate early vs delayed EN improves clinical and nutrition outcomes, although the definitions of early EN and late EN differ across the studies.^{4–6} In 688 children with $\geq 10\%$ total body surface area burns, those randomly assigned to early EN within 3–6 h from injury had significantly lower mortality compared with children who received delayed EN after 48 h.⁴ A smaller study randomly assigned 77 children with $\geq 25\%$ total body surface area burns to early EN within 24 h of burn vs EN delayed after 48 h.⁶ Compared with children with delayed EN, those receiving early EN had significantly reduced caloric deficits during postburn weeks 1 and 2, higher insulin and triiodothyronine levels, and decreased 3-methylhistidine levels in week 1, possibly suggesting an anabolic

TABLE 1 Timing of enteral nutrition initiation.

Reference	Study design; number of sites	Study aims	Patient characteristics (n)	Intervention	Outcomes	Comments
<i>RCTs</i>						
Prakash et al ³	Open-label RCT; single center	To compare the effect of early initiation of EN ≤ 6–24 h vs late EN > 24 h on length of PICU stay	n = 120 n = 60 early EN n = 60 late EN	Randomized to receive early EN vs late EN Median time to start EN was 6 h in the early group and 40 h in the late group	No differences in PICU length of stay or other clinical outcomes No adverse events with early EN	Study lacked adequate power to determine outcomes of early vs late EN Baseline differences in demographics with higher prevalence of malnutrition in the early EN group may confound outcomes
Khorasani et al ⁴	RCT; single center	To evaluate the safety and effectiveness of early EN ≤ 3–6 h of injury vs late EN > 48 h in pediatric patients with ≥ 10% TBSA burn injuries	n = 688 n = 366 early EN Mean (SD) age, 5 (3.5) years n = 322 late EN Mean (SD) age, 5 (3) years	Randomized to receive early EN or late EN	Early- compared with late-EN group had reduced mortality ($P < 0.05$), decreased length of hospital stay ($P < 0.05$), and less weight loss ($P < 0.05$)	Actual timing of EN initiation was not reported Findings may not apply to the heterogeneous population of children with critical illness
Venter et al ⁵	RCT; single center	To investigate the clinical outcomes with early EN ≤ 24 h after burn injury vs late EN > 48 h during the resuscitation phase in children with ≥ 20% TBSA burn injuries	n = 18 n = 9 early EN Median age 4.54 years n = 9 late EN Median age, 4.45 years	Randomized to receive early EN vs late EN Median (range) time to start EN was 10.7 h (5.5–23 h) in the early group and 54 h (51–58 h) in the late EN group	Early- compared with late-EN group had higher insulin ($P = 0.008$) and insulin to glucagon ratios ($P = 0.043$), suggesting an anabolic response No differences in clinical outcomes	Small sample size limits ability to detect between-group differences Findings may not apply to population of children with critical illness
Gottschlich et al ⁶	RCT; single center	To examine the effectiveness and safety of early EN ≤ 24 h vs late EN > 48 h in pediatric patients with burn injuries > 25% TBSA	n = 72 n = 36 early EN Mean (SD) age, 9.1 (0.7) years n = 36 late EN Mean (SD) age, 9.6 (0.7) years	Randomized to receive early EN vs late EN EN was initiated in mean (SD) 15.6 h (1 h) in the early group and 48.5 h (0.4 h) in the late group ($P < 0.0001$) EN met goal in mean (SD) 30.3 h (2.9 h) in the early group vs 50.3 h (2.1 h) in the late group ($P < 0.0001$)	Early- compared with late-EN group had reduced caloric deficits during postburn weeks 1 ($P < 0.0001$) and 2 ($P = 0.0022$), higher insulin ($P = 0.0004$) and triiodothyronine levels ($P = 0.0162$), and decreased 3-methylhistidine levels in week 1 ($P = 0.0138$)	No clinical outcome differences Nonstatistical increase in adverse events in early-EN group, with intestinal infarction in 5 patients causing 4 deaths Authors note possibility of weak anabolic response with changes in hormones and 3-methylhistidine

TABLE 1 (Continued)

Reference	Study design; number of sites	Study aims	Patient characteristics (n)	Intervention	Outcomes	Comments
<i>Observational studies</i>						
Powell et al ⁷	Retrospective study; single center	To investigate the association of nutrition support practices with outcomes in infants and children with ARDS requiring mechanical ventilation Early EN defined as achievement of 25% of cumulative energy goal within 48 h	n = 151 Median (IQR) age, 4.1 (0.63–12.55) years	Compared daily EN and PN delivery with nutrition goals for 7 days following ARDS diagnosis Energy needs assessed with estimated resting energy expenditure multiplied by stress factor	Early EN vs non-early EN was associated with reduced mortality (adjusted OR, 0.071; 95% CI, 0.009–0.542; P = 0.011), shorter PICU length of stay (adjusted RR, 1.79; 95% CI, 1.25–2.55; P = 0.001), and more ventilator-free days (median, 13.8 vs 19.9 days; P = 0.017)	Actual timing of EN initiation was not reported No EN guideline for standardization of nutrition support practice Potential for overfeeding with use of stress factors applied to resting energy expenditure Retrospective, single-center design with restriction to ARDS diagnosis limits generalizability
Solana et al ⁸	Secondary analysis of prospective observational multicenter study; 12 PICUs	To define barriers to early EN in children with critical illness and to analyze differences in energy and protein delivery, complications, and outcomes between early and late EN Early vs late EN analyzed independently as ≤24 h vs >24 h and ≤48 h vs >48 h; different comparisons to avoid overlap	n = 68 n = 15 EN ≤ 24 h Median (IQR) age, 12 (2–32 months) n = 53 EN > 24 h Median (IQR) age, 8 (3–43) months n = 46 EN ≤ 48 h Median (IQR) age, 5 (2–18) months n = 22 EN > 48 h Median (IQR) age, 36 (8–134) months	Outcomes and complications were analyzed with initiation of EN ≤ 24 h vs >24 h, and ≤48 h vs >48 h	≤48-h early EN compared with >48-h late EN was associated with higher mean energy intake (P = 0.04), faster attainment of maximum energy provision (P < 0.01), and less constipation (P = 0.01) ≤24-h early EN vs >24-h late EN was associated with shorter duration of mechanical ventilation (P = 0.04)	Definite timing of EN initiation was not reported Severity of illness was positively correlated with the timing of EN initiation (P = 0.02) Small sample size limits ability to assess outcomes Potential for overfeeding confounding outcomes with rapid attainment of estimated energy requirements
Shahi et al ⁹	Retrospective study; single center	To evaluate outcomes of children with ≥10% TBSA burn injuries who receive early EN < 4 h from admission vs late EN ≥ 4 h via gastric vs postpyloric tube	n = 132 n = 63 early EN Mean (SD) age, 4.35 (4.62) years n = 69 late EN Mean (SD) age, 4.25 (3.99) years n = 68 oral nutrition n = 45 EN via NGT n = 19 EN via postpyloric tube	Nutrition and clinical outcomes were compared between patients receiving early and late EN and between gastric and postpyloric EN route	Early-EN group had lower rates of underfeeding during the first week (P = 0.014) and shorter PICU length of stay (P = 0.025)	Actual timing of EN initiation was not reported No difference in pneumonia adjusted for number of ventilator days with NGT vs postpyloric EN Outcomes limited by retrospective design and small sample size

(Continues)

TABLE 1 (Continued)

Reference	Study design; number of sites	Study aims	Patient characteristics (<i>n</i>)	Intervention	Outcomes	Comments
Srinivasan et al ¹⁰	Secondary analysis of HALF-PINT multicenter RCT; 31 PICUs	To determine clinical outcomes associated with early EN ≤ 48 h of HALF-PINT study randomization vs no early EN (EN > 48 h or no EN) in children with hyperglycemia requiring inotropic support and/or mechanical ventilation	<i>n</i> = 608 <i>n</i> = 331 early EN <i>n</i> = 277 no early EN	Nutrition support guideline provided to participating sites Collected and analyzed clinical and demographic data	Early EN compared with no early EN was associated with lower 90-day hospital mortality ($P = 0.007$), more ventilator-free days ($P = 0.003$), less organ dysfunction ($P < 0.001$), more PICU-free days ($P = 0.02$), and more hospital-free days ($P = 0.003$)	Definite timing of EN initiation was not reported Lack of nutrition support data between PICU admission and HALF-PINT study randomization Adherence to nutrition support guideline likely varied across sites Secondary analysis not powered to detect clinical outcomes associated with EN timing
Balakrishnan et al ¹¹	Retrospective multicenter study; 5 PICUs	To examine EN practices in children with TBI and to evaluate risk factors associated with delayed EN (> 48 h) Early EN defined as ≤ 48 h	<i>n</i> = 416 <i>n</i> = 347 early EN <i>n</i> = 69 late EN	Compared demographics, injury and procedure data, time to initiation of nutrition support, and injury and illness severity scores between patients who received early EN and delayed EN	Lower Glasgow Coma Scale scores and higher Injury Severity Score were associated with late EN (both $P < 0.0001$) Late EN was associated with worse functional status at PICU discharge ($P = 0.02$)	Actual timing of EN initiation was not reported Retrospective design limits ability to interpret all data affecting outcomes Research in children with TBI may not be generalizable to children with critical illness
Bagci et al ¹²	Prospective multicenter database study; 9 PICUs	To evaluate the impact of early EN (< 24 h) on mortality and factors associated with the timing of EN initiation and attainment of the EN target (25% of cumulative energy goal ≤ 48 h) Late EN was defined as > 24 h	<i>n</i> = 95 Mean (SD) age, 4 (4.7) years <i>n</i> = 45 early EN <i>n</i> = 20 early EN ≤ 6 h <i>n</i> = 25 early EN ≤ 7 –24 h <i>n</i> = 43 late EN > 24 h <i>n</i> = 7 no EN	Obtained demographic, clinical, and nutrition information from database to analyze outcomes with early EN and target EN	Early EN was associated with lower illness severity score ($P = 0.012$) Achievement of target EN was associated with reduced 28-day mortality risk (adjusted OR, 0.06; 95% CI, 0.01–0.47; $P = 0.007$)	Actual timing of late EN initiation was not reported Lower illness severity may allow for greater feasibility of early and adequate EN
Haney et al ¹³	Retrospective database study; single center	To investigate the impact of early EN (80% estimated energy or protein needs within 72 h of PICU admission) in patients	<i>n</i> = 106 <i>n</i> = 58 early EN Median (IQR) age, 4 (1.0–17.0) months	Clinical, demographic, and nutrition data including daily energy and protein intake were obtained from database and	Early EN compared with no early EN was associated with decreased PICU (adjusted $P = 0.001$) and hospital length of stay (adjusted $P < 0.001$)	Definite timing of EN initiation was not reported Retrospective study design limits ability to

TABLE 1 (Continued)

Reference	Study design; number of sites	Study aims	Patient characteristics (n)	Intervention	Outcomes	Comments
Mikhailov et al ¹⁴	Retrospective database multicenter study; 2 PICUs	with acute respiratory failure requiring mechanical ventilation	n = 48 no early EN Median (IQR) age, 25.5 (2.0–144.5) months	compared with outcomes between children who received early EN vs no early EN	Early EN vs no early EN in the 2 PICUs was associated with lower total (adjusted $P = 0.0006$) and daily (adjusted $P < 0.001$) hospital charges	interpret all variables impacting outcomes Potential risk for overfeeding with intake of 80% resting energy expenditure in <72 h
		To determine the association between early EN (25% of cumulative energy goal ≤ 48 h) and PICU and hospital charges in children with critical illness	n = 859 n = 171 early EN n = 688 no early EN	Hospital charges were analyzed in patients receiving early EN vs no early EN	Actual timing of EN initiation was not reported Interaction between PICU sites and early EN possibly due to differences in patient population or EN practices Retrospective study design limits ability to collect all data impacting outcomes	
Mikhailov et al ¹⁵	Retrospective multicenter database study; 12 PICUs	To evaluate the association of early EN (25% of cumulative energy goal ≤ 48 h) with mortality and morbidity in children with PICU length of stay ≥ 96 h	n = 5105 n = 1386 early EN Median (IQR) age, 0.7 (0.3–2.9) years n = 3719 no early EN Median (IQR) 4.0 (0.8–11.6) years	Demographic, clinical, and nutrition data were obtained from database and outcomes were analyzed in patients receiving early EN vs no early EN	Early EN vs no early EN was associated with lower mortality (adjusted OR, 0.51; 95% CI, 0.34–0.76; $P = 0.001$), and no difference in duration of mechanical ventilation or length of stay	Definite timing of EN initiation was not reported Lower severity-of-illness score with early EN vs no early EN ($P < 0.0001$) suggests greater feasibility of early EN in patients who are less sick Retrospective study design limits ability to obtain all data impacting outcomes
Lopez-Herce et al ¹⁶	Prospective observational study; single center	To evaluate EN tolerance in children with critical illness and shock vs without shock	n = 526 n = 65 with shock Median (IQR) age, 12 (0.7–264) months n = 461 without shock Median (IQR) age, 5 (0.1–228) months	EN formula type, duration, tolerance, and GI complications were compared in patients receiving postpyloric EN with shock vs without shock	Median (range) time to the start of EN was 2 (0.1–28) days in the shock group and 2 (0.4–90; $P = 0.431$) days in the group without shock Higher incidence of GI complications ($P = 0.0001$), abdominal distension/gastric residue ($P = 0.004$),	Increased mortality in children with shock vs without shock ($P = 0.0001$) 1 death in a child with shock was related to mechanical complication of EN with duodenal perforation

(Continues)

TABLE 1 (Continued)

Reference	Study design; number of sites	Study aims	Patient characteristics (n)	Intervention	Outcomes	Comments
Sanchez et al ¹⁷	Prospective observational study; single center	To compare tolerance of transpyloric EN started early (≤24 h of PICU admission) vs late (>24 h) in children with critical illness	n = 526 Mean (SD) age, 23 (41) months n = 202 EN < 24 h n = 324 EN > 24 h in median (range) 3 (1–43) days	Compared energy intake, tolerance, and complications of early EN and late EN	Early EN compared with late EN was associated with less abdominal distention (P = 0.05) and no differences in diarrhea or clinical outcomes	Actual timing of early EN initiation was not reported 3 patients in the late EN group developed necrotizing enterocolitis
Sanchez et al ¹⁸	Prospective observational study; single center	To assess the feasibility of transpyloric EN (early defined as ≤48 h) in children after cardiac surgery vs other children with critical illness	n = 350 n = 212 following cardiac surgery n = 138 other PICU patients	Compared nutrition support data and gastrointestinal complications with transpyloric EN in children after cardiac surgery vs other children with critical illness	Mean (SD) time to start EN in the cardiac surgery group was 5.4 (8.6) days and 3.9 (4.3) days in non-cardiac surgery group (P < 0.05) Late initiation of EN was associated with pulmonary infections (P = 0.03) No association between GI complications and early vs late transpyloric EN	2 patients postoperative from cardiac surgery developed necrotizing enterocolitis

Abbreviations: ARDS, acute respiratory distress syndrome; CI, confidence interval; EN, enteral nutrition; GI, gastrointestinal; HALF-PINT, Heart and Lung Failure-Pediatric Insulin Titration Trial; IQR, interquartile range; NGT, nasogastric tube; OR, odds ratio; PICU, pediatric intensive care unit; PN, parenteral nutrition; RCT, randomized controlled trial; RR, risk ratio; TBI, traumatic brain injury; TBSA, total body surface area.

response with reduced protein catabolism. Significantly reduced rates of underfeeding during postburn week 1 were also demonstrated in a retrospective observational study of 132 children with $\geq 10\%$ total body surface area burn injuries when EN was initiated within 4 h vs after 4 h.⁹ Research in children with burn injuries demonstrating clinical and nutrition outcome benefits with early vs delayed EN may not be applicable to the more heterogeneous population of children with critical illness who present with a variety of diagnoses and comorbid conditions.

Although no further randomized controlled trials of early vs delayed EN on the impact of outcomes are available in this population, nonrandomized multicenter studies are important to review. In a secondary analysis of the Heart And Lung Failure Pediatric Insulin Titration (HALF-PINT) study of tight glycemic control with insulin in children with hyperglycemia requiring inotropic support and/or mechanical ventilation, differences in mortality with early vs delayed EN or no EN were evaluated.¹⁰ A nutrition support guideline emphasizing early EN was provided to all participating sites. A total of 608 children from 31 PICUs were included, and 54% received early EN within 48 h of HALF-PINT study randomization. Early EN was significantly associated with lower 90-day hospital mortality, more ventilator-free days, less organ dysfunction, more PICU-free days, and more hospital-free days vs children receiving late EN or no EN. Nutrition support data were not available prior to randomization. Although multiple variables were controlled in the statistical analyses, including severity-of-illness score and PICU site, it is possible that some children able to receive early EN were less sick than those with delayed or no EN and may confound these results. Adherence to the nutrition support guideline likely varied across PICU sites, with the potential for underfeeding and overfeeding impacting outcomes. Although not powered to detect clinical outcomes associated with the timing of EN initiation, this large observational study is generalizable to current PICU practice.

Early initiation of EN may result in quicker achievement of higher energy intakes during pediatric critical illness. In an additional secondary analysis of a multicenter observational study including 68 infants and children from 12 PICUs in Spain, EN started within 48 h vs after 48 h was associated with significantly higher energy intake and faster attainment of maximum energy provision,⁸ guided by the Schofield equation.¹⁹ With EN initiated in 24 h compared with after 24 h, a significantly shorter duration of mechanical ventilation was revealed, although severity of illness was significantly and positively correlated with the timing of EN initiation. In this small study, early EN within 24 h or within 48 h

was likely more feasible in patients with a lower illness severity, thus contributing to a reduced dependency on respiratory support and allowing for possible overfeeding with rapid attainment of estimated energy requirements. In a small retrospective study of children with critical illness, the impact of early exclusive EN was examined on clinical outcomes.¹³ Of the 106 patients included, 55% met $\geq 80\%$ of estimated resting energy expenditure based on the World Health Organization equation²⁰ and 80% of protein intake goals according to 2009 ASPEN guidelines²¹ within 72 h of admission, defined as early EN. Children in the early EN group received a significantly higher percentage of energy and protein intake targets during PICU admission vs those who did not receive early EN. There were no differences in the duration of mechanical ventilation; however, PICU length of stay was significantly shorter in the children who received early EN. The rapid achievement of 80% of estimated energy and protein targets by day 3 in this small study may have increased the risk of overfeeding and the potential for adverse outcomes. Furthermore, the retrospective study design prevented control of all variables impacting outcomes.

The dose response to early EN, defined as a cumulative intake of $\geq 25\%$ of goal calories, as assessed for individual patients according to bedside clinician practice, within 48 h has been investigated by several authors. In a retrospective multicenter study of 5105 children with critical illness and length of stay ≥ 96 h in the PICU, early EN was associated with significantly lower odds of death.¹⁵ Results remained significant after adjustment for propensity score, severity of illness, age, and center. A much smaller multicenter prospective observational study of 9 PICUs in Turkey evaluated early EN in 95 children with critical illness.¹² Achievement of 25% of cumulative goal calories within 48 h was found to be a significant predictor of reduced 28-day mortality risk. Similar findings were reported in a small single-center retrospective study of 151 infants and children with acute respiratory distress syndrome requiring mechanical ventilation.⁷ Early EN providing 25% of cumulative goal calories within 48 h was associated with a significant reduction in PICU mortality after adjustment for age and severity of disease. Observational studies do not determine cause and effect, and statistical methodologies are unable to completely control for all variables impacting study outcomes. Children receiving early EN within 24–48 h are often less sick than children with delayed or no EN; however, these observational studies contribute important data on feasibility and efficacy of early EN.

Outcome benefits associated with early EN must be balanced with the risks of this strategy. In some studies

of infants and children with critical illness and severe burns, congenital heart disease, or shock, abdominal distention and diarrhea are reported with early initiation of EN, and less frequently, severe complications, including necrotizing enterocolitis and intestinal infarction, are described.^{6,16,18} Published nutrition support guidelines for infants and children with critical illness recommend EN with stable vasoactive support,^{1,2} and a cautious approach with close monitoring of gastrointestinal tolerance to EN is warranted in this population.

Despite uncommon reports of adverse events, early EN in 24–48 h from admission or injury is safe and tolerable for the majority of children with critical illness. Early EN improves gastrointestinal tolerance in some patients, with reports of decreased constipation⁸ and reduced abdominal distension.¹⁷ Results of these observational studies are confounded by the inability to control for all variables impacting study outcomes, and more randomized controlled trials are needed to determine safety, tolerance, and efficacy of early vs delayed EN.

The timing of EN initiation within 24–48 h vs delayed after 48 h has demonstrated outcome benefits in infants and children with critical illness as shown in Figure 1. Providing early vs delayed EN is significantly associated with reduced mortality,^{7,10,12,15} less organ dysfunction,¹⁰ reduced ventilator requirement,^{7,8,10} decreased length of stay in the intensive care unit,^{7,13} and hospital,¹³ cost savings,¹⁴ and attainment of energy and protein intake

goals.^{8,13} Conversely, late EN initiation >48 h after admission was significantly associated with pulmonary infections in a prospective observational study of 350 infants and children with critical illness in whom the majority were admitted following surgery for congenital heart disease.¹⁸ In addition, a retrospective multicenter study of 416 children with traumatic brain injury demonstrated significant associations between delayed initiation of EN of >48 h and lower Glasgow Coma Scale scores, higher Injury Severity Score, and worse functional status at discharge.¹¹

Data demonstrating outcomes with early EN within 24–48 h vs delayed after 48 h were derived mainly from observational cohorts with heterogeneous populations in which variables were not completely controlled. The definition of early EN differed across studies, and much of the available research is fraught by low statistical power because of a small number of included patients. Retrospective collection of data limits the ability to evaluate all pertinent information that can be obtained prospectively. It is paramount to emphasize that observational research suggesting improved outcomes with the timing of EN in <24–48 h compared with EN delayed after 48 h may reflect a greater feasibility of providing EN to infants and children with lower severity of critical illness.

TIMING OF ACHIEVEMENT OF TARGET EN INTAKE

The timing from admission to the PICU to the achievement of a specific targeted EN intake has the potential to impact outcomes as summarized in Table 2. In a multicenter prospective study of nutrition practices in 500 children with critical illness from 31 PICUs, EN was initiated within 48 h in 60% of patients.²⁶ Cumulative mean EN intake >66.6% of energy goal vs 33.3%–66.6% or <33.3% of target, as assessed for individual patients according to bedside clinician practice, up to the first 10 days in the PICU was significantly associated with reduced mortality. Subsequently, in a prospective multicenter observational study of 1245 children with critical illness from 59 PICUs, EN was initiated by PICU day 2 in 60% of patients.²⁵ Compared with cumulative mean EN intake of <20% of protein goal within the first 10 days in the PICU, intakes of 20%–60% and >60% of protein goal, as assessed according to bedside clinician practice, were significantly associated with reduced mortality. With a low reported use of indirect calorimetry and methods to assess protein balance by the individual centers, the majority of energy and protein targets were determined through different methods of estimation, thus increasing the risk of overfeeding, underfeeding, and confounding

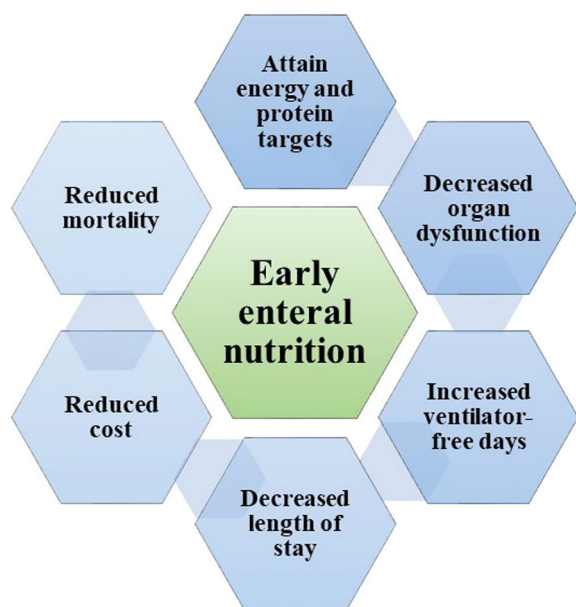


FIGURE 1 Outcome benefits associated with early enteral nutrition. Enteral nutrition initiation within 24–48 h of admission facilitates timely attainment of energy and protein intake goals and is associated with improved clinical outcomes, including mortality, ventilator-free days, and length of stay.

TABLE 2 Timing of achievement of target enteral nutrition intake.

Reference	Study design; number of sites	Study aims	Patient characteristics (n)	Intervention	Outcomes	Comments
<i>Observational studies</i>						
Zaher et al ²²	Prospective observational study; single center	To assess delivered vs estimated needs of EN energy and macronutrients, and to examine the association of macronutrient intake and systemic inflammation in children with critical illness requiring mechanical ventilation	n = 87 Median (IQR) age, 2 (0.6–4.9) years	EN intake was recorded and macronutrient intake was compared with estimated requirements Anthropometric measurements were obtained, and inflammatory cytokine levels were measured	Protein ($P = 0.01$) and energy intake ($P = 0.04$) were negatively correlated with increased IL-6 on days 2–3 Increased fat intake was associated with elevated TNF- α on days 4–8 in children >1 year of age ($P = 0.032$)	No association was found between energy or macronutrient intake and clinical outcomes Small, single-center study lacked adequate power to determine clinical outcome differences between macronutrient intake
Velazco et al ²³	Prospective observational multicenter study; 57 PICUs	To investigate energy and protein delivery to PICU patients requiring mechanical ventilation and awaiting or following a surgical procedure	n = 519 Median (IQR) age, 2 (0.5–8) years	Recorded adequacy of cumulative energy and protein delivery, with <60% of the prescribed protein goal established as an inadequate intake	Inadequate vs adequate EN protein delivery was associated with longer time to EN initiation ($P < 0.001$) and longer duration of EN interruptions ($P < 0.001$)	Authors suggest an increased EN administration rate to account for interruptions, and continuation of EN through procedures if secured airway and EN via postpyloric route
Canarie et al ²⁴	Retrospective observational multicenter study; 6 PICUs	To examine EN practices to determine risk factors associated with delayed EN (> 48 h) in children with critical illness Early EN defined as started in ≥ 48 h from PICU admission	n = 444 Median (IQR) age, 4.0 (0.5–11.9) years n = 356 early EN Median (IQR) age, 4.1 (0.5–12.3) years n = 88 delayed EN Median (IQR) age, 4.2 (0.8–11.1) years	Collected data on prescribed and delivered EN, respiratory support, procedures, and GI disturbances Patients were divided into early-EN group and delayed-EN group	Delayed EN was associated with higher levels of invasive ($P = 0.02$) and noninvasive ventilation ($P = 0.001$), severity-of-illness score ($P = 0.001$), procedures ($P = 0.001$), and longer time to meet the EN goal ($P < 0.001$)	Retrospective study design does not allow for collection of all data impacting outcomes Study not powered to determine clinical outcomes Estimation of energy and protein needs increases the risk of underfeeding and overfeeding
Mehta et al ²⁵	Prospective observational multicenter study; 59 PICUs	To examine the association between protein intake and 60-day mortality in children requiring mechanical ventilation	n = 1245 Median (IQR) age, 1.7 (0.4–7.0) years	Calculated daily and cumulative mean adequacies of energy and protein delivery as a percentage of prescribed goal for up to 10 days, and examined association of	Adequacy of EN protein intake was associated with 60-day mortality (adjusted $P < 0.001$) Compared with EN protein intake <20% goal, 20%–60% goal (OR, 0.37;	Most patients had estimations of energy and protein requirements Study restricted to ≥ 8 -bed PICUs and children requiring mechanical ventilation

(Continues)

TABLE 2 (Continued)

Reference	Study design; number of sites	Study aims	Patient characteristics (<i>n</i>)	Intervention	Outcomes	Comments
Mehta et al ²⁶	Prospective observational multicenter study; 31 PICUs	To examine factors influencing the adequacy of energy and protein intake in the PICU and to describe their relationship to clinical outcomes in children requiring mechanical ventilation	<i>n</i> = 500 Mean (SD) age, 4.5 (5.1) years	Calculated daily and cumulative mean adequacies of energy and protein delivery as a percentage of prescribed goal for up to 10 days Patients were followed for up to 60 days or until hospital discharge	Compared with cumulative mean EN intake <33.3% prescribed, 33.3–66.6% of energy goal (OR, 0.27; 95% CI, 0.11–0.67; <i>P</i> = 0.002) and >66.7% goal was associated with reduced mortality risk (OR, 0.14; 95% CI, 0.03–0.61; <i>P</i> = 0.002) Use of estimated energy requirements may increase risk of underfeeding and overfeeding Severity-of-illness scores were missing in 31% of patients Study restricted to ≥8-bed PICUs and children requiring mechanical ventilation	Early initiation, postpyloric route, shorter interruptions, larger PICU size, and a dedicated PICU dietitian were associated with higher EN protein delivery 95% CI, 0.17–0.76; <i>P</i> = 0.007) and >60% goal (OR, 0.14; 95% CI, 0.04–0.52; <i>P</i> = 0.003) were associated with reduced 60-day mortality risk
Briassoulis et al ²⁷	Prospective observational study; single center	To determine if early achievement of energy and protein balance improves nitrogen balance and creatinine height index during critical illness in children requiring mechanical ventilation	<i>n</i> = 71 Ages, 2–204 months	All patients started EN within 12 h of admission, with energy intake increasing to 150% predicted BMR by day 5 Nitrogen balance and creatinine height index were obtained on days 1 and 5	All patients had negative nitrogen balance and moderate to severely low creatinine height index on day 1 Protein intake was positively correlated with nitrogen balance (<i>P</i> < 0.001) and increased energy intake (<i>P</i> < 0.003) Patients with positive vs negative nitrogen balance on day 5 had higher protein adequacy (<i>P</i> < 0.05)	Increased risk for overfeeding with energy intake of 150% BMR by day 5

Abbreviations: BMR, basal metabolic rate; CI, confidence interval; EN, enteral nutrition; GI, gastrointestinal; IL-6, interleukin-6; IQR, interquartile range; OR, odds ratio; PICU, pediatric intensive care unit; TNF- α , tumor necrosis factor alpha.

of outcomes. These multicenter studies were also restricted to children requiring mechanical ventilation and PICU size of at least eight beds, limiting wider applicability of the findings. Severity-of-illness scores were not available for all enrolled patients and may not reflect actual illness severity or expected patient outcomes. In addition, the observations of reduced mortality with the timing of achievement of specific EN targets are associations and do not infer causality.

Nutrition outcomes may also improve with achievement of a target EN intake. Nitrogen balance was examined prospectively in a small study of 71 children requiring mechanical ventilation.²⁷ EN initiated within 12 h of admission reached a target intake of 150% of predicted basal metabolic rate by day 5. Although all patients had a negative nitrogen balance on day 1, protein and energy intakes were significantly higher in patients attaining positive nitrogen balance on day 5 compared with children with negative nitrogen balance. Achievement of 150% estimated energy goal by day 5 in this study suggests a significant component of overfeeding with an increased risk for associated adverse outcomes. Observational studies with estimates of energy and protein requirements are unable to adequately control for all variables impacting nutrition outcomes in children with critical illness, and randomized trials are needed.

Delaying the initiation of EN >48 h may negatively impact the ability to meet estimated energy and protein intake goals. Examination of protein intake in a prospective multicenter cohort study of 519 patients with critical illness requiring mechanical ventilation and awaiting or following a surgical procedure demonstrated delayed EN initiation was associated with protein intake deficits.²³ Median protein intake in 66% of patients receiving EN was <15% of target and 67% received an insufficient cumulative protein intake, defined as <60% of goal during the first 7 days of admission. Inadequate protein intake was significantly associated with a longer time to EN initiation. EN delayed 48 h after admission was significantly associated with failure to meet prescribed energy and protein intakes in a retrospective multicenter study of 444 children in 6 PICUs.²⁴ An increase in circulating levels of inflammatory cytokines was also revealed with suboptimal protein intake in a prospective observational study of 87 children with critical illness receiving EN.²² These prospective and retrospective observational studies provide important insights into potential outcomes associated with delayed EN but are unable to completely exclude all variables that confound the reported outcomes. Using estimates of energy and protein intake goals increases the risk for overfeeding and underfeeding, adversely impacting

outcomes. Children with higher illness severity may be more likely to have delayed initiation of EN and subsequently reduced opportunities for the timely achievement of energy and protein intake targets.

The timing of achievement of target EN intake in infants and children with critical illness has been associated with reduced mortality^{25,26} and positive nitrogen balance.²⁷ These outcomes have been exclusively evaluated in observational studies and primarily with estimation of energy and protein targets. Early and aggressive efforts to meet estimated caloric and protein goals may increase the risk of overfeeding and subsequent adverse outcomes in infants and children with critical illness.²⁸ More precise and feasible measures of protein balance and energy expenditure, including compatibility with different modes of respiratory support and extracorporeal devices are needed to more accurately determine outcomes associated with the timing of achievement of target EN intake.

TIMING OF PN INITIATION WITHOUT CONTRAINDICATIONS TO EN

The current SCCM-ASPEN recommendation regarding the timing of PN initiation¹ is based on the Early versus Late Parenteral Nutrition in the Pediatric Intensive Care Unit (PEPaNIC) randomized controlled multicenter trial.²⁹ The PEPaNIC study compared outcomes of children provided early PN within the first 24 h of PICU admission to those assigned to late PN on day 8 of admission if EN was <80% of goal. A total of 1440 term infants and children up to 17 years of age with critical illness, a score of ≥ 2 on the Screening Tool Risk on Nutritional Status and Growth (STRONGkids),³⁰ and no contraindications to EN from three PICUs were randomly assigned to early or delayed PN. A STRONGkids score of 0 indicates low risk of malnutrition, 1–3 suggests medium risk, and 4–5 implies high risk.³⁰ Initiation of EN was attempted early in both groups and was increased in accordance with local guidelines. All patients received intravenous micronutrients from day 2 until EN provided 80% of energy targets. Intravenous micronutrient dosing practice varied between the centers.³¹ PN dose and composition also varied and was used to supplement EN to meet estimated energy and macronutrient needs as determined by the individual centers. Children randomly assigned to early vs late PN had significantly higher rates of new infection, increased duration of mechanical ventilation, and longer PICU length of stay, and a higher number of patients required renal replacement therapy.

No statistical difference in mortality was shown between groups receiving early vs late PN.

The PEPaNIC randomized controlled trial, several of the preplanned secondary analyses, and other studies evaluating the timing of PN initiation on outcomes are summarized in Table 3. Outcomes reported with late vs early PN in the PEPaNIC secondary analyses include cost reduction,⁴² decreased infections in patients diagnosed with malnutrition by anthropometric measurements,⁴³ less refeeding hypophosphatemia,³² decreased total thyroxine levels,³⁹ and no difference in weight-for-age Z score deterioration.³⁸ An association of increased infections and prolonged mechanical ventilation requirement was found with early exposure to PN amino acids but not with lipid or dextrose.⁴⁵ Secondary analyses of the PEPaNIC study also reported findings of long-term follow-up of development,^{34,37,40} quality of life,³⁶ and alterations to DNA.^{33,35,44}

In the PEPaNIC randomized controlled trial, delayed supplemental PN initiation resulted in superior outcomes compared with early PN.²⁹ Although early PN was the standard of care in the centers participating in the PEPaNIC trial, initiation of PN within 24 h of PICU admission is not consistent with typical practice. The accuracy of the STRONGkids screening tool to detect risk of malnutrition, used for eligibility for patient inclusion in the PEPaNIC study, has been questioned,¹ as it has not been validated in children with critical illness.⁴⁶ Also, children identified with malnutrition according to baseline anthropometric measurements or unable to receive EN were underrepresented in the trial.⁴⁷ Glucose control and nutrition support practices were not uniform between centers, and the definitions of infection were not standardized. The likelihood of overfeeding with PN initiation within 24 h of PICU admission using estimates of energy requirements with concurrent attempts to increase EN may have contributed to adverse clinical outcomes.⁴⁸ Ultimately, more research is necessary to recommend the optimal timing of initiation of supplemental and exclusive PN for infants and children with critical illness.

Additional studies have explored outcomes associated with the timing of supplemental PN without contraindications to EN. A secondary analysis of a randomized controlled multicenter trial of therapeutic hypothermia in 90 children with severe traumatic brain injury from 15 centers demonstrated initiation of EN or PN within 72 h was significantly associated with survival and favorable 6- and 12-month disposition.⁴¹ The adequacy of EN and PN intake was unable to be determined despite prospective data collection, and the results of children with traumatic brain injury may not be generalizable to those with critical illness. Nosocomial infections were

retrospectively examined in 204 children with critical illness receiving nutrition support.⁴⁹ Patients were divided into two groups: EN only and EN with supplemental PN. Early supplemental PN was defined as initiation within 48 h and late was described as started between days 2 and 6. Supplemental PN compared with EN only was associated with a significant increase in acquired infections. The time overlap of early and late PN may limit the ability to discern outcome differences between groups in this study. With the use of estimated energy requirements, overfeeding may have confounded the observed outcomes. In addition, significantly more children with higher illness severity received supplemental PN compared with those with EN only, possibly contributing to the increased morbidity. Observational investigations, particularly when retrospective in design, are limited by multiple uncontrolled variables impacting outcomes.

TIMING OF ACHIEVEMENT OF TARGET INTAKE WITH EN AND PN

In the absence of randomized trials assigning children with critical illness to the initiation of EN vs PN, large observational studies may have greater external validity to inform nutrition support interventions in the PICU. A recent prospective observational study provides a more representative description of PICU nutrition practices in 1844 children with critical illness requiring mechanical ventilation from 77 PICUs located in 17 countries around the world.²⁸ The median adequacy from EN and PN was 49% of goal calories and 66% of goal protein. Energy and protein goals were assessed for individual patients according to local site practice. Mortality significantly decreased in proportion to increased energy and protein intake adequacy. On day 8, a minimum of 60% of the EN goal was met by 62% of patients for energy and 59% of patients for protein, whereas $\geq 60\%$ of combined EN and PN goal was achieved by 73% for energy and 71% for protein in the entire cohort. Patients who received a minimum of 60% of energy and protein targets by day 3, defined as early, and by day 7, classified as pragmatic, had significantly lower mortality compared with those meeting $<60\%$ of goals, with no significant difference in mortality between the early and pragmatic groups. Significantly fewer children in the most severe illness category met a minimum of 60% of energy and protein goals from exclusive EN, but this was not seen with combined EN and PN. Patients who met 60% adequacy from exclusive EN or combined EN and PN by day 7 demonstrated significantly lower hazards of 60-day

TABLE 3 Timing of parenteral nutrition initiation without contraindications to enteral nutrition.

Reference	Study design; number of sites	Study aims	Patient characteristics (n)	Intervention	Outcomes	Comments
<i>RCTs</i>						
Fivez et al ²⁹	RCT, multicenter study; 3 PICUs Investigators not aware of treatment assignments	To evaluate primary end point of new infection and duration of PICU dependency in patients receiving early (first 24 h) vs late PN (day 8)	n = 1440 Children with critical illness, term neonates to children 17 years of age n = 723 early PN Median (IQR) age, 1.4 (0.3–6.1) years n = 717 late PN Median (IQR) age, 1.5 (0.2–7.2) years	Randomized to receive early or late PN until EN met at least 80% goal, with early EN attempted in both groups, and IV micronutrients provided to both groups	Patients receiving early vs late PN had higher rates of new infections (18.5% vs 10.7%; adjusted $P < 0.001$), increased duration of mechanical ventilation (6.4 ± 0.7 days vs 4.4 ± 0.3 days; adjusted $P = 0.001$), longer PICU stay (9.2 ± 0.8 days vs 6.5 ± 0.4 days; adjusted $P < 0.001$), higher prevalence of renal replacement therapy (3.6% vs 2.5%; adjusted $P = 0.04$), and no statistical difference in 90-day mortality (6.8% vs 5.3%; adjusted $P = 0.08$)	Early PN was the standard of care at all participating centers STRONGkids screening tool to assess malnutrition risk used as inclusion criteria for the study not validated for critically ill children Nutrition support practices varied between centers Infection definition not standardized PN within 24 h of PICU admission is not typical practice Potential risk for overfeeding with early PN and concurrent attempts to increase EN
<i>Observational studies</i>						
Veldscholte et al ³²	Observational study, secondary analysis of PEPaNIC multicenter RCT; 3 PICUs	To assess the occurrence of early RFH and association with clinical outcomes in patients receiving early (first 24 h) vs late PN (day 8)	n = 1247 (patients from PEPaNIC study) n = 618 early PN n = 629 late PN Early RFH defined as serum phosphate < 0.65 mmol/L with a decline of > 0.16 mmol/L	Serum phosphate concentrations were compared between early and late PN	Early PN was associated with higher incidence of early RFH vs late PN (78% vs 22%, $P < 0.001$) Early RFH associated with increased length of PICU ($P = 0.003$) and hospital stay ($P = 0.007$)	Overall incidence of early RFH was relatively low at 3% Patients in the late-PN group who discharged from the PICU prior to day 8 did not receive PN, possibly confounding outcomes

(Continues)

TABLE 3 (Continued)

Reference	Study design; number of sites	Study aims	Patient characteristics (n)	Intervention	Outcomes	Comments
Jacobs et al ³³	Observational study, preplanned secondary analysis of PEPaNIC multicenter RCT; 3 PICUs	To investigate if DNA methylation changes in patients who received early (first 24 h) vs late PN (day 8) can explain negative impacts on emotion and behavior	within the first 3 days of PICU admission n = 403 (patients from PEPaNIC study) n = 192 early PN n = 211 late PN	DNA methylation and reported behavior data compared between early and late PN	Early PN was associated with altered methylation of 37 CpG sites in leukocyte DNA	Statistically provides biological basis for reported long-term adverse effect of early vs late PN on behavioral and emotional development Unclear if short-term PN exposure can independently cause short- and long-term adverse effects
Verlinden et al ³⁴	Observational study, secondary analysis of PEPaNIC multicenter RCT; 3 PICUs	To determine the role of patient age at time of exposure to early (first 24 h) vs late PN (day 8) on the impact of long-term, 2-year neurocognitive development	n = 786 (patients from PEPaNIC study) n = 121 age ≤28 days (56 early PN, 65 late PN) n = 239 age 29 days to <11 months, (120 early PN, 119 late PN) n = 223 age 11 months to <5 years (110 early EN, 113 late PN) n = 203 age ≥5 years (105 early PN, 98 late PN)	Physical and neurocognitive development was assessed 2 years after the PEPaNIC study, results were analyzed by age group and early vs late PN	Children aged 29 days to <11 months at the time of critical illness receiving early PN were the most vulnerable to poor developmental outcomes	Authors suggest none of the age groups received developmental benefit from early PN provision Unclear whether short-term PN exposure can independently cause described long-term adverse effects
Guiza et al ³⁵	Observational study, preplanned	To investigate DNA methylation	n = 825 (patients from PEPaNIC study)	DNA methylation of patients in the PEPaNIC	159 CpG sites were methylated differently in patients vs controls	Authors propose DNA methylation of specific CpG sites

TABLE 3 (Continued)

Reference	Study design; number of sites	Study aims	Patient characteristics (n)	Intervention	Outcomes	Comments
Hordijk et al ³⁶	secondary analysis of PEPaNIC multicenter RCT; 3 PICUs	changes in children with critical illness, and assess the effect of early (first 24 h) vs late PN (day 8) on DNA methylation changes	n = 403 early PN n = 411 late PN n = 352 healthy controls	study with a blood sample from the last day of PICU admission receiving early vs late PN was compared with matched healthy controls	Early PN and amino acid dose contributed to the different methylation of 37 (23%) of the CpG sites	may explain the effect of early PN and early PN amino acids on neurocognitive development reported at 2 years Unclear if short-term PN exposure can independently cause long-term adverse effects
	Observational study, secondary analysis of PEPaNIC multicenter RCT; 3 PICUs	To examine the impact of early (first 24 h) vs late PN (day 8) on HRQOL of patients and parents 2 years after PEPaNIC RCT vs healthy matched controls	n = 786 (patients from PEPaNIC study) n = 391 early PN n = 395 late PN n = 405 healthy controls	Parents completed HRQOL questionnaires based on the age of their child, and surveys intended to assess HRQOL of the parents	Parents of patients reported lower HRQOL scores for their children vs those of healthy matched controls No significant difference in HRQOL between early- and late-PN groups	Patients have worse reported HRQOL vs healthy children, but no differences were found in HRQOL between early vs late PN
Jacobs et al ³⁷	Observational study, secondary analysis of PEPaNIC multicenter RCT; 3 PICUs	To evaluate the effects of early (first 24 h) vs late PN (day 8) on physical, neurological, behavioral, and emotional development 4 years after PEPaNIC RCT	n = 684 (patients from PEPaNIC study) n = 328 early PN n = 356 late PN n = 369 healthy controls	Neurocognitive development was assessed at 4 years following the PEPaNIC RCT	Compared with controls, patients had lower height ($P < 0.0001$), head circumference ($P = 0.00077$), and intelligence quotient ($P < 0.0001$) and worse health status ($P < 0.0001$), neurocognitive function ($P < 0.0001$), and behavioral developmental outcomes ($P = 0.00055$) Late or no PN was protective against emotional and behavioral problems ($P = 0.0070$)	Patients have worse development vs healthy children, and late PN may protect against emotional and behavioral problems observed in children who received early PN Unclear whether short-term PN exposure can independently cause reported long-term adverse effects

(Continues)

TABLE 3 (Continued)

Reference	Study design; number of sites	Study aims	Patient characteristics (n)	Intervention	Outcomes	Comments
van Puffelen et al ³⁸	Observational study, secondary analysis of PEPaNIC multicenter RCT; 3 PICUs	To assess the effect of late PN (day 8) vs early PN (first 24 h) on weight-for-age Z score	n = 470 (patients from PEPaNIC study) n = 236 early PN n = 234 late PN	Effect of early vs late PN on change in weight-for-age Z score, and association between weight-for-age Z score change and outcomes was analyzed	Less weight Z score deterioration was associated with lower risk of new infection (adjusted OR per Z score increase, 0.72; 95% CI, 0.55–0.96; <i>P</i> = 0.02) and higher likelihood of earlier live discharge from the PICU (adjusted HR per Z score increase, 1.22; 95% CI, 1.10–1.37; <i>P</i> < 0.001)	No significant difference in weight-for-age Z score change with early vs late PN Infection definition not standardized
Jacobs et al ³⁹	Observational study, preplanned secondary analysis of PEPaNIC multicenter RCT; 3 PICUs	To investigate the prognostic value of NTI and evaluate the impact of late (day 8) vs early PN (first 24 h) on NTI	n = 982 (patients from PEPaNIC study) n = 64 (healthy controls)	Serum TSH, T4, T3, and rT3 levels of children with critical illness were compared with those of healthy controls	Lower serum TSH, T3, T4, and T3/rT3 levels and higher T3 levels in population with critical illness (<i>P</i> < 0.0001), associated with increased morbidity and mortality (all <i>P</i> < 0.0001) Late PN further decreased T4 levels	Laboratory samples drawn from heparinized lines prevented assessment of free hormone concentration in patients with critical illness Decrease in T4 with late PN may be detrimental Decreased T3/rT3 with late PN suggests beneficial inactivation of peripheral thyroid hormone
Verstraete et al ⁴⁰	Observational study, secondary analysis of PEPaNIC multicenter RCT; 3 PICUs	To examine the long-term developmental outcomes of patients who received early (first 24 h) vs late PN (day 8)	n = 786 (patients from PEPaNIC study) n = 391 early PN n = 395 late PN n = 405 healthy controls	Physical and neurocognitive development assessed 2 years after PEPaNIC study and in healthy controls	No differences in anthropometrics, health status, or neurological outcomes with early vs late PN Late PN was associated with better inhibitory control (adjusted <i>P</i> = 0.0001) and executive functioning (adjusted <i>P</i> = 0.011) vs early PN	Young patient age precluded some neurocognitive testing Neuroimaging studies not done Unclear whether short-term PN exposure can independently

TABLE 3 (Continued)

Reference	Study design; number of sites	Study aims	Patient characteristics (n)	Intervention	Outcomes	Comments
Meinert et al ⁴¹	Secondary analysis of “the Cool Kids Trial” multicenter RCT; 15 PICUs	To examine the relationship between timing of initiation of EN or PN in children with severe TBI and clinical outcomes	n = 90 Age, <18 years	The timing of EN and PN initiation was collected, categorized, and compared with clinical outcomes	Initiation of EN or PN ≤ 72 h after TBI was associated with decreased mortality ($P = 0.01$) and improved functional outcomes at 6 months ($P = 0.03$) and 12 months ($P = 0.04$)	cause long-term adverse effects Patients in the late-PN group who left PICU prior to day 8 did not receive PN Investigators were unable to determine the amount or adequacy of EN and PN Results may not be generalizable to children during critical illness without TBI
van Puffelen et al ⁴²	Observational study preplanned secondary analysis of the PEPaNIC multicenter RCT; 2 PICUs	To describe the cost-effectiveness of late (day 8) vs early PN (first 24 h) in 2 of the 3 PEPaNIC study sites	n = 1343 (patients from PEPaNIC study) n = 670 early PN n = 673 late PN	Hospital costs of groups receiving early vs late PN were compared	Mean direct medical costs were 21% lower ($P = 0.007$) for patients receiving late vs early PN	Higher costs for early-PN group attributed to increased length of PICU stay and costs associated with treating nosocomial infections Infection definition not standardized
van Puffelen et al ⁴³	Observational study, secondary analysis of PEPaNIC multicenter RCT; 3 PICUs	To assess the effects of late (day 8) vs early PN (first 24 h) in patients with critical illness identified with undernourishment diagnosed by admission anthropometric	n = 289 (patients from PEPaNIC study) n = 139 early PN n = 150 late PN Undernourishment defined as weight-for-age or BMI-for-age Z score < -2 in children <1 year	Infections and PICU length of stay assessed in children with undernourishment receiving late vs early PN	Late vs early PN was associated with 11% decreased risk of new infections (adjusted OR, 0.39; 95% CI, 0.19–0.78; $P = 0.01$) and median 2-day reduction in PICU stay (adjusted HR, 1.37; 95% CI, 1.06–1.75; $P = 0.01$) No significant differences between groups for mortality, hypoglycemia, weight deterioration	Nonstandard definition of malnutrition Infection definition not standardized STRONGkids malnutrition screening tool not validated in children with critical illness, and may have affected the number (Continues)

TABLE 3 (Continued)

Reference	Study design; number of sites	Study aims	Patient characteristics (<i>n</i>)	Intervention	Outcomes	Comments
Verstraete et al ⁴⁴	Observational study; preplanned secondary analysis of PEPaNIC multicenter RCT; 3 PICUs	To examine telomere length of children with critical illness and determine whether early (first 24 h) vs late PN (day 8) affects telomere length	<i>n</i> = 1148 (patients from PEPaNIC study) <i>n</i> = 342 healthy controls	Leukocyte telomere length of children with critical illness was compared with those of healthy controls Telomere length of patients receiving early vs late PN was compared	Leukocyte telomeres were 6% shorter than normal in children with critical illness Early PN was associated with worsening telomere shortening vs late PN (<i>P</i> < 0.0001)	of patients identified with undernourishment Patients in the late-PN group who left the PICU prior to day 8 did not receive PN Further study is needed to determine if shortened telomere length predisposes patients to poor outcomes Unclear if short-term PN exposure can independently cause long-term adverse effects
Vanhorebeek et al ⁴⁵	Observational study secondary analysis of PEPaNIC multicenter RCT; 3 PICUs	To analyze early macronutrient provision impact on new infections, weaning from mechanical ventilation, and PICU length of stay in patients receiving early (first 24 h) vs late PN (day 8)	<i>n</i> = 1440 (patients from PEPaNIC study)	Percentage of macronutrients provided in PN were assessed to examine associations between individual macronutrients and outcomes	Higher vs lower amino acids dose was associated with higher likelihood of acquiring a new infection (adjusted HR per 10% increase 1.043–1.134 for days 1–5; <i>P</i> < 0.029), decreased likelihood of earlier live weaning from mechanical ventilation (HR, 0.950–0.975 for days 3–7; <i>P</i> < 0.045), decreased likelihood of early live discharge from the PICU (HR, 0.943–0.972 for days 1–7; <i>P</i> < 0.030) Higher vs lower glucose provision was associated with fewer infections (HR, 0.870–0.913 for days 1–3; <i>P</i> < 0.036) Higher vs lower lipid administration was associated with earlier PICU discharge (HR, 1.027–1.050 for days 4–7; <i>P</i> < 0.043)	Infection definition not standardized Study suggests early and increased doses of amino acids, not lipids or dextrose, are associated with adverse outcomes PN amino acid dose not randomized Maximal risk of harm with amino acid doses <1.5 g/kg and much lower doses also associated with adverse outcomes Severity-of-illness scores were controlled in the

TABLE 3 (Continued)

Reference	Study design; number of sites	Study aims	Patient characteristics (<i>n</i>)	Intervention	Outcomes	Comments
Wang et al ⁴¹	Observational, retrospective study; single center	To investigate the relationship between supplemental PN and nosocomial infections and the effect of early (first 48 h) vs late supplemental PN (days 2–6) on new infection rates Nosocomial infections classified by national diagnostic criteria	<i>n</i> = 204 Median (IQR) age, 9.0 (3.0–30) months <i>n</i> = 101 (EN only) <i>n</i> = 103 (supplemental PN) <i>n</i> = 60 (early supplemental PN) <i>n</i> = 43 (late supplemental PN)	The effect of early vs late supplemental PN on nosocomial infections was compared with EN only	Supplemental PN compared with EN only was associated with an increase in nosocomial infections (34.0% vs 10.9%, <i>P</i> < 0.001) Late supplemental PN was an independent predictor of nosocomial infection (OR, 3.40; 95% CI, 1.13–10.19; <i>P</i> = 0.029)	Data from retrospective single-center studies are limited by multiple uncontrolled variables analyses; however, these scores may not represent actual illness severity Patients with perceived increased illness severity may have been prescribed higher PN amino acid doses, possibly confounding results

Abbreviations: BMI, body mass index; CI, confidence interval; CpG, 5'-cytosine-phosphate-guanine-3'; EN, enteral nutrition; HR, hazard ratio; HRQOL, health-related quality of life; IQR, interquartile range; IV, intravenous; NTI, nonthyroidal illness; OR, odds ratio; PEPaNIC, Early versus Late Parenteral Nutrition in the Pediatric Intensive Care Unit; PICU, pediatric intensive care unit; PN, parenteral nutrition; RCT, randomized controlled trial; RFH, refeeding hypophosphatemia; rT3, reverse triiodothyronine; STRONGkids, Screening Tool for Risk on Nutritional Status and Growth; T3, triiodothyronine; T4, total thyroxine; TBI, traumatic brain injury; TSH, thyrotropin.

mortality compared with patients who did not meet 60% goals by day 7, defined as the late group. The incidence of infection and duration of ventilator-free days was not affected by the attainment of 60% of energy and protein intake goals. Achievement of 60% energy and protein targets in 3 days was not superior to reaching goals in 7 days, which is more practical and allows for a less aggressive approach. In contrast to other studies that focus on outcomes related to the initiation of nutrition support, this study demonstrated that an intake of 60% cumulative energy and protein goal by PICU day 3 or 7 is significantly associated with reduced mortality. Results of observational studies must be interpreted with caution, as associations do not imply cause and effect. Children with a higher severity of illness may receive delayed nutrition support because of the priority of other urgent interventions, potentially confounding the results of these observations.

The timing and quantity of supplemental PN in children with traumatic brain injury receiving inadequate EN and the impact on clinical outcomes have been evaluated in small observational studies. A secondary analysis of a randomized controlled multicenter trial of therapeutic hypothermia in 90 children with severe traumatic brain injury from 15 centers demonstrated initiation of EN or PN within 72 h was significantly associated with survival and favorable 6- and 12-month disposition.⁴¹ In a retrospective observational investigation of 109 children with traumatic brain injury, EN or PN was started in a median of 1.5 days and met target in a median of 3.4 days.⁵⁰ Both early initiation of nutrition support within 72 h and full caloric intake by day 7 were significantly and positively correlated with reduced intensive care unit length of stay. Attainment of energy target by day 7 was also significantly associated with an improved disposition at hospital discharge. Energy targets were based on needs in healthy children in this study and likely exceeded requirements, increasing the risk of overfeeding and confounding of outcomes. Furthermore, children with a lower injury severity likely had fewer competing priorities of care, allowing for earlier initiation of EN or PN, compared with those who were more severely injured. Multiple factors may impede initiation and adequate delivery of nutrition support in children with traumatic brain injury, and research in this subgroup may not be generalizable to other populations with critical illness.

Outcomes associated with the attainment of energy and protein goals have also been investigated in small studies of children with acute respiratory failure in the PICU. A retrospective study of 107 children with acute respiratory distress syndrome demonstrated significantly reduced mortality with energy intake $\geq 80\%$ of predicted

resting energy expenditure using the Schofield equation¹⁹ and minimum protein intake of 1.5 g/kg by day 3 compared with patients who did not achieve these targets.⁵¹ Conversely, protein intake of ≥ 1.5 g/kg on PICU day 3 was associated with significantly prolonged duration of mechanical ventilation and PICU length of stay in a retrospective study of 74 infants and young children with critical illness secondary to bronchiolitis.⁵² These small studies demonstrating opposing outcomes with reference to 1.5 g protein/kg intake lack sufficient power to determine differences between groups. The retrospective study designs also increase the risk of multiple uncontrolled variables, including the possibility of overfeeding impacting outcomes.

Studies evaluating the timing of achievement of target intake with EN and PN and associated outcomes in children with critical illness are reviewed in Table 4. The available evidence is limited to observational cohorts receiving supplemental PN. In the largest of these studies, a minimum intake of 60% of estimated energy and protein requirements by day 7 was significantly associated with reduced mortality.²⁸ Observational studies of children with critical illness receiving nutrition support may be more generalizable compared with randomized controlled trials evaluating a specific subgroup or interventions, but as such are limited by variables that cannot be completely controlled. Children who are less ill are likely able to receive earlier and increased amounts of EN, thus avoiding the use of supplemental PN. More research representing typical bedside practice is required to determine the optimal timing and dose of supplemental PN when EN is inadequate to improve outcomes in this population.

TIMING OF PN INITIATION WHEN EN IS CONTRAINDICATED

At present, no study has randomly assigned children with critical illness to receive early vs delayed PN (or no nutrition support) when EN is contraindicated or restricted to a minimal intake. Observational studies, although with inherent limitations, can provide relevant data on the impact of timing of PN as the main source of nutrition support that is not otherwise available in randomized trials and are reviewed in Table 5. A prospective multicenter observational study of 500 children requiring mechanical ventilation from 31 PICUs demonstrated significantly higher mortality in patients receiving PN.²⁶ A significant association between early PN and mortality was revealed in patients without EN intake for the first 4 days of admission in a retrospective multicenter study of 2069 children from 12 PICUs.⁵⁴

TABLE 4 Timing of achievement of target intake with enteral nutrition and parenteral nutrition.

Reference	Study design; number of sites	Study aims	Patient characteristics (n)	Intervention	Outcomes	Comments
<i>Observational studies</i>						
Bechard et al ²⁸	Prospective multicenter observational study; 77 PICUs	To examine the association between the time to achieve delivery of 60% of the prescribed energy and protein targets and clinical outcomes in children requiring mechanical ventilation	n = 1844 Median (IQR) age, 1.64 (0.47–7.05) years	Daily energy and protein delivery from EN and PN were recorded for up to 10 days Patients were categorized based on time to reach 60% prescribed energy and protein and analyzed for associations with clinical outcomes	Cumulative 7-day intake ≥60% vs <60% prescribed energy ($P = 0.007$) and protein goal ($P = 0.027$) from EN or EN and PN was associated with reduced 60-day mortality, with no differences in acquired infections or ventilator-free days	Large observational studies may be more representative of typical bedside practice vs RCTs, although associations do not infer cause and effect Children with lower severity of disease may be more likely to receive an increased EN or PN adequacy
Ng et al ⁵²	Retrospective observational study; single center	To describe nutrition status and nutrition support practices in children with severe bronchiolitis and to explore associations with outcomes Underfeeding defined as <80% and overfeeding as >120% Estimated energy needs Adequate protein intake defined as >1.5 g/kg	n = 74 Median (IQR) age, 4.9 (2.0–10.4) months	Clinical, demographic data, weight-for-length Z scores, energy and protein intake vs estimated requirements for the first 3 days of PICU admission were recorded and analyzed for associations with outcomes	Weight-for-length Z score <−2 was associated with prolonged duration of PICU stay ($P = 0.028$) and mechanical ventilation ($P = 0.004$) Greater protein adequacy >1.5 g/kg on PICU day 3 was associated with longer PICU stay ($P = 0.009$) and more ventilator days ($P = 0.015$)	No association between timing of EN initiation, overfeeding or underfeeding with PICU or mechanical ventilation dependency The retrospective study design and use of estimated energy and protein needs increase the risk for uncontrolled variables impacting outcomes
Wong et al ⁵¹	Retrospective observational study; single center	To describe EN and PN received by children with ARDS and to determine if provision of an adequate energy (≥80% of predicted resting energy expenditure) and protein intake (≥1.5 g/kg/day) by day 3 is associated with improved clinical outcomes	n = 107 Median (IQR) age, 5.2 (1.0–10.4) years	Compared patients that received adequate vs inadequate energy and protein intake and clinical outcomes	Initiation of EN in <24 h ($P = 0.008$) and adequate vs inadequate energy ($P = 0.025$) and protein ($P = 0.002$) were associated with reduced mortality Adequate vs inadequate protein intake was associated with increased ventilator-free days (adjusted $P = 0.044$)	Overfeeding vs. no overfeeding was associated with reduced mortality ($P = 0.004$) and was not associated with extubation failure The retrospective study design and assessment of energy and protein with estimations of needs increase the variability and confounding of outcomes

(Continues)

TABLE 4 (Continued)

Reference	Study design; number of sites	Study aims	Patient characteristics (<i>n</i>)	Intervention	Outcomes	Comments
Taha et al ⁵⁰	Retrospective study; single center	To examine the timing of and EN or PN initiation and attainment of goal energy intake in relation to ICU length of stay and disposition status at hospital discharge in children with TBI	<i>n</i> = 109 Median (range) age, 13 (8–18) years	Collected data on time of EN or PN initiation and time to reach prescribed EN or PN goal and compared with clinical outcomes	Median (range) time to EN or PN initiation was 1.5 (0.02–11.9) days, and time to energy goal was 3.4 (0.5–19.6) days Initiation of EN or PN ≤ 72 h (<i>P</i> < 0.01) and attainment of energy goal by day 7 were associated with reduced ICU length of stay (<i>P</i> < 0.01) Energy goal by day 7 was associated with better disposition status at hospital discharge (<i>P</i> < 0.0001)	Energy targets based on needs for healthy children, increasing the risk of overfeeding and confounding of outcomes Results may be limited to specific population of children with TBI and not generally applicable to other children with critical illness

Abbreviations: ARDS, acute respiratory distress syndrome; EN, enteral nutrition; ICU, intensive care unit; IQR, interquartile range; PICU, pediatric intensive care unit; PN, parenteral nutrition; RCT, randomized controlled trial; TBI, traumatic brain injury.

TABLE 5 Timing of PN initiation when EN is contraindicated.

Reference	Study design; Number of sites	Study aims	Patient characteristics (n)	Intervention	Outcomes	Comments
<i>Observational studies</i>						
Ariagno et al ⁵³	Prospective, observational cohort study; single center	To describe PN use in the PICU, specifically looking at timing of initiation, delivery of macronutrients, and effect on weight-for-age Z scores Early PN defined as initiation within 48 h from admission to PICU Late PN defined as provision at 5 days Short PN duration defined as <5 days	n = 95 Median (IQR) age, 5.6 (0.2–45.9) months with PICU stay >3 days	Timing of PN initiation and PN duration was categorized Provision of macronutrients and change in weight-for-age Z score was calculated	33% of patients received early PN Median (IQR) PN duration was 8 (5–14) days 16.8% of patients received PN for short duration Median (IQR) energy (29% [3%–50%] vs 55% [40%–74%]; $P < 0.001$) and protein intakes were lower (31% [4%–47%] vs 72% [44%–81%]; $P < 0.001$) with late vs early PN A 0.50-unit decline in weight-for-age Z score was observed with late PN (95% CI, 0.11–0.89; $P = 0.012$)	Late PN provision at day 5 was significantly associated with lower adequacy of macronutrient provision and a greater decline in weight-for-age Z score Findings from small, single-center observational cohort studies are limited by multiple uncontrolled variables
Goday et al ⁵⁴	Retrospective, observational multicenter study; 12 PICUs	To assess if early PN (providing 25% of energy goals within 48 h of PICU admission) and no EN for the first 96 h of PICU admission were associated with poor outcomes	n = 2069 Median (IQR) age, 6.61 (1.64–12.92) years	Patient characteristics, duration of mechanical ventilation, mortality, and nutrition provision data were obtained from the participating PICUs and analyzed	Early PN was associated with mortality in patients without EN intake for the first 4 days of admission (OR, 2.10; 95% CI, 1.41–3.13; $P = 0.0003$)	No differences were seen in PICU length of stay or duration of mechanical ventilation Data from retrospective studies are limited by the inability to control all variables
Anderson et al ⁵⁵	Pre/post quasi-experimental study; single center	To implement a process to delay PN initiation for up to 5 days following PICU admission	n = 2333 (preintervention) Median (IQR) age, 3 (0–11) years n = 2491 (postintervention) Median (IQR) age, 3 (1–12) years	Dietitian and pharmacist jointly wrote PN orders, and provided ongoing education to staff	5.5% of patients received PN prior to day 5 before vs 3.1% after intervention ($P < 0.001$) PICU patients receiving PN for ≤ 3 days decreased from 2.6% to 1.5% before and after intervention, respectively ($P = 0.01$)	No significant differences in PICU or hospital length of stay before and after intervention Authors suggest a delay in PN initiation until day 5 of PICU admission does not adversely affect length of stay Interventions not randomized

(Continues)

TABLE 5 (Continued)

Reference	Study design; Number of sites	Study aims	Patient characteristics (n)	Intervention	Outcomes	Comments
Armstrong et al ⁵⁶	Retrospective, observational cohort study; single center	To assess predictors of energy and protein delivery in the first 2 weeks of ECMO therapy in the PICU Early PN defined as PN started within 48 h of PICU admission Late PN defined as PN initiation after 48 h of PICU admission	n = 54 Median (IQR) age, 0.1 (0–16.4) months	Demographic and clinical data were collected for patients who survived 24 h after ECMO cannulation	51 (94%) patients received PN, 40 (74%) early PN, 11 (22%) late PN Late compared with early PN was associated with protein deficits (7.9 ± 3.8 g/kg vs 5.1 ± 3.3 g/kg; <i>P</i> = 0.019) Younger age (<i>P</i> = 0.0098) and venoarterial ECMO mode (<i>P</i> = 0.0014) were associated with decreased likelihood of EN initiation	No difference in energy deficit with late vs early PN Authors suggest PN is indicated during weeks 1–2 of ECMO support to prevent protein and energy intake deficits Findings from small, single-center retrospective studies are limited by multiple uncontrolled variables
Greathouse et al ⁵⁷	Retrospective, observational cohort study; single center	To evaluate the effect of the timing of EN initiation on survival for pediatric patients requiring ECMO	n = 49 Mean (SD) age, 53 (76) months	Demographic and clinical data were collected, and nutrition support data were reviewed on day 5 of ECMO support Patients receiving PN were compared with patients receiving any EN on day 5	Kaplan-Meier curves indicated a greater survival to discharge with any EN exposure on day 5 vs exclusive PN on day 5 (<i>P</i> = 0.031) Total energy intake on day 5 was associated with decreased mortality risk (HR, 0.20; 95% CI, 0.06–0.71; <i>P</i> = 0.013)	Patients who received nutrition support closer to goal energy provision, and those who received any EN had improved survival to discharge Data from small, retrospective cohort studies are limited by the inability to control all variables
Mehta et al ²⁶	Prospective, observational multicenter cohort study; 31 PICUs	To examine factors affecting adequate provision of energy and protein in the PICU, as well as the relationship of provision adequacy to outcomes in children requiring mechanical ventilation	n = 500 Mean (SD) age, 4.5 (5.1) years	Nutrition practices were recorded for a maximum of 10 days of PICU stay Patients were followed for up to 60 days or until hospital discharge	67% of patients received EN, 8.8% PN, 21% EN and PN Provision of PN was associated with increased mortality risk (OR, 2.6; 95% CI, 1.3–5.3; <i>P</i> = 0.008)	Prescribed energy and protein goals were based on estimated requirements in most patients Findings from observational studies are limited by multiple uncontrolled variables

TABLE 5 (Continued)

Reference	Study design; Number of sites	Study aims	Patient characteristics (n)	Intervention	Outcomes	Comments
Pettignano et al ⁵⁸	Retrospective, observational cohort study; single center	To compare tolerance, complications, and adequacy of nutrition support provision between patients undergoing ECMO and receiving either EN or PN	$n = 27$ $n = 14$ received PN Median (range) age, 21.5 (1.5–168) months $n = 13$ received EN Median (range) age, 72 (2–192) months	Demographic and clinical data were collected	No difference in time to meet energy goal (3.07 ± 2.1 days PN vs 4.25 ± 2.6 days EN; $P = 0.54$), mortality, or sepsis goal between groups Direct costs were 5 times higher for PN vs EN	Interventions not randomized Data from small, retrospective, single-center studies are limited by multiple uncontrolled variables

Abbreviations: CI, confidence interval; ECMO, extracorporeal membrane oxygenation; EN, enteral nutrition; HR, hazard ratio; IQR, interquartile range; OR, odds ratio; PICU, pediatric intensive care unit; PN, parenteral nutrition.

Early PN was defined as exclusive PN meeting a cumulative 25% goal energy during the first 48 h of admission. No differences were seen in PICU length of stay or duration of mechanical ventilation. These observations are limited by an inability to completely control all variables impacting outcomes, and caution is required when interpreting and applying results to bedside practice. It is likely that children unable to receive EN had increased severity of illness and the additional contribution of PN exposure to the observed outcomes cannot be determined.

A variety of practices are used to provide PN to children with critical illness when EN intake is inadequate. Some are driven by international or national guidelines and others by local or institutional protocol. A survey examining PN practices in 81 PICUs internationally found that 43 had read the PEPaNIC study.⁵⁹ Of those 43 respondents, 14% reported continuing to initiate PN within the first 24 h of PICU admission, 23% within 48 h, 40% within the first week, and 23% withheld PN until day 8 of admission.

Although many centers have incorporated findings of the PEPaNIC study into nutrition support protocols, different approaches have been used, and some have closely examined practices regarding the timing of PN initiation. A single-center pre/post quasi-experimental study investigated the effects of PN initiation delayed until day 5 of PICU admission.⁵⁵ Included were 2333 patients prior to the intervention and 2491 after the practice change that received PN and were admitted to the PICU within the first 24 h of hospital admission with minimum age of 30 days. Improvements were demonstrated in the avoidance of early PN before day 5, as well as a decrease in short-term PN for ≤ 3 days, without change to PICU or hospital length of stay. Investigators assessed that the lack of difference in length of stay suggested the practice of delaying PN initiation did not adversely affect outcomes. Although changes in PN initiation before and after the change in practice were clinically and statistically significant, a much larger population may be necessary to demonstrate clinical outcome differences. A single-center prospective study of PN initiation in the PICU including 95 children showed median time to PN initiation was 4 days.⁵³ Investigators found 33% of patients received early PN defined as within 48 h from admission to the PICU. Median PN duration was 8 days, and in 16.8% of children, PN was given for a short duration specified as < 5 days. Median percentage adequacies of energy and protein provision were significantly lower, and a significant decline in weight-for-age Z score was observed in patients with PN initiation after 48 h from PICU admission, designated as late PN, vs children receiving early PN. Small single-center studies

inform quality improvement initiatives for local PICU practice but may not be generalizable to other centers. Results of these studies are in contrast to those from the PEPaNIC trial demonstrating a shorter length of stay with delayed PN²⁹ and no difference in deterioration of weight-for-age Z score with early vs late PN.³⁸

The timing of PN initiation when EN intake is minimal or contraindicated is variable and often based on clinician judgment and the clinical and nutrition status of individual patients. A single-center retrospective study examined the timing of PN and ability to achieve protein and energy delivery goals by day 7 in 54 pediatric patients who received extracorporeal membrane oxygenation (ECMO).⁵⁶ PN was provided to 94% of patients during the first 7 days of ECMO; 9% received only PN, whereas 85% received a combination of EN and PN. Of the 51 patients who received PN, 74% received early PN within 48 h of admission and 22% received late PN provided after 48 h. Late PN was significantly associated with cumulative protein deficits and inability to meet calorie and protein goals by day 7 compared with the early group. An additional cohort of pediatric patients undergoing ECMO and receiving PN was retrospectively examined to determine the impact of time related EN exposure on outcomes.⁵⁷ EN and PN intakes were evaluated in 49 children on day 5 of ECMO support. Compared with those receiving exclusive PN on day 5, children exposed to any EN on day 5 demonstrated a significantly greater survival to discharge. Total energy intake on day 5, whether from PN or PN and EN, was also significantly associated with improved survival. Research findings from observational studies, particularly retrospective in design, are limited by multiple uncontrolled variables and should be interpreted with caution. Furthermore, outcomes observed in subgroups, such as children requiring ECMO, may not apply to other children with critical illness.

The timing of PN initiation and attainment of energy and protein goals, as a supplement to inadequate EN or as the main source of nutrition, has the potential to affect nutrition and clinical outcomes in infants and children with critical illness. The available literature is limited by study design and small sample sizes, and controversy remains regarding the optimal time of initiation, EN or PN route of administration, and timing of achievement of a specific dose of energy and protein. Although external validity of the PEPaNIC trial and the secondary analyses may be limited, the novel research provides important insights and hypotheses on the timing of PN initiation in the PICU.⁶⁰ Remaining investigations regarding the timing of PN initiation and achievement of target intake in this population are restricted to observational studies. Although observational studies may be more generalizable, they lack

the ability to control multiple variables that confound outcomes. Observations can be biased by participant selection in which patients who are less sick receive earlier and higher amounts of EN and demonstrate improved clinical outcomes vs those who are more ill and have delayed provision of EN and PN.

PRACTICAL STRATEGIES FOR TIMING NUTRITION SUPPORT IN THE PICU

The timing and adequacy of EN and PN provision impact important nutrition and clinical outcomes in infants and children with critical illness. Consequently, it is essential to implement guidelines and recent evidence into practice.⁶¹ Figure 2 presents several strategies for timing nutrition support interventions in the PICU. A nutrition support team or multidisciplinary group including a dietitian, nurse, pharmacist, and physician can assess institution-specific barriers and facilitate creation and implementation of local nutrition support protocols to optimize the delivery of EN and PN.^{1,2} Documentation of energy requirement in the medical record was examined in a multicenter retrospective study of 5 PICUs including



FIGURE 2 Practical Strategies for Timing Nutrition Support Interventions in the PICU. Strategies to improve enteral nutrition delivery in the PICU include multidisciplinary teamwork, objective assessment of gastrointestinal tolerance, use of protein-energy-enriched formulas and an enteral nutrition protocol, and regularly auditing practice. GI, gastrointestinal; PICU, pediatric intensive care unit.

patients between 30 days and 18 years of age admitted over a 2-year period.⁶² Of 1369 patients reviewed, 47.7% had evidence of energy requirement recorded and were significantly more likely to receive EN on days 1–4 of admission compared with patients without documented assessment of caloric need. As the energy requirement was recorded in the medical record by a registered dietitian 95.6% of the time, this study highlights the importance of timely assessment of energy requirements by a dedicated PICU dietitian.

Barriers to adequate nutrition support therapy in the PICU outlined in Table 6 must be addressed and individualized by each center to optimize energy and protein intake and to improve patient outcomes.⁶⁹ Common barriers to EN and PN initiation and achievement of an adequate energy and protein intake are related to patient condition, clinician priorities, and gastrointestinal tolerance to EN.^{63,66–68} Feeding intolerance was recently investigated in a systematic review of 31 publications, including 2973 pediatric patients with critical illness.⁷⁰ Definitions of intolerance included discontinuation of EN, gastrointestinal symptoms, and inability to reach the EN goal. Median prevalence of feeding intolerance was 20%, most commonly because of large gastric residuals, abdominal distention, diarrhea, and vomiting. Feeding intolerance was associated with reduced energy delivery, delay in meeting the EN target, pulmonary infections, sepsis, and mortality.

Assessment of gastrointestinal tolerance to EN has historically included evaluation of intragastric fluid during continuous EN or in between bolus feedings.

The practice of gastric residual volume measurement was evaluated in 87 children with critical illness by comparing patients from a PICU with routine measurement of gastric residuals to those from a PICU without this practice.⁷¹ Patients without gastric residual monitoring had significantly increased severity-of-illness scores on admission, and significantly longer durations of mechanical ventilation and PICU stay. Other differences included a significantly shorter time from admission to EN initiation and a lower incidence of continuous EN modality in the group measuring vs not measuring gastric residuals. No significant difference was observed in the median percentages of energy goals achieved between groups. Despite the higher illness severity in patients without measurement of gastric residuals, the incidence of vomiting, necrotizing enterocolitis, and ventilator-associated pneumonia was not statistically different between groups.

Gastrointestinal tolerance to EN may be improved with a modification to the formula composition, a change in mode or route of administration, or the use of medications to treat constipation and delayed gastric motility.⁷² Maximizing the benefits and tolerance of EN for patients with critical illness may also be aided with use of an EN protocol, as reviewed in Table 7, although these outcomes require multidisciplinary support and routine use of the guideline. EN algorithms were only used by 9 of 31 PICUs in a multicenter study and use had no association with EN adequacy.⁷⁴

A systematic review of nine studies including 1564 children with critical illness assessed the quality of outcome data of using a PICU EN guideline with the

TABLE 6 Reported barriers to nutrition support in the pediatric intensive care unit.

Barriers to enteral nutrition	Barriers to parenteral nutrition
Intubation/extubation ^{58,63–65}	Fluid restriction ^{63,67,68}
Procedures/surgery/imaging ^{24,58,63–68}	Hyperglycemia ⁶³
Hemodynamic/respiratory/medical instability ^{58,63,64,66,67}	Intravenous access/intravenous device issues ⁶³
Gastrointestinal intolerance ^{24,58,63,65,67,68}	Medication administration ⁵⁸
Enteral access/feeding tube issues ^{63,65,66}	Hemodynamic instability ⁶³
Fluid restriction ^{63,67,68}	Transfers between units ⁶³
Illness severity ^{8,24}	
Invasive mechanical ventilation ^{8,24}	
Noninvasive positive pressure ventilation ²⁴	
Opioid requirement ⁸	
Medication administration ⁵⁸	
Upper gastrointestinal bleeding ⁶⁷	
Red blood cell transfusion ⁵⁸	
Hyperglycemia ⁶³	
Lack of enteral nutrition guideline ⁶⁴	
Lack of dietitian presence ⁶⁶	
Provider knowledge deficit ⁶⁶	
Transfers between units ⁶³	

TABLE 7 Enteral nutrition algorithms.

Reference	Study design; number of sites	Study aims	Patient characteristics (n)	Intervention	Outcomes	Comments
<i>Observational studies</i>						
Rungsattatharm et al ⁷³	Retrospective study; single center	To investigate the impact of a nutrition support guideline on attaining energy and protein targets in patients admitted to the PICU	n = 215 n = 53 preintervention Median (IQR) age, 14 (4–104) months n = 162 postintervention Median (IQR) age, 16 (6–69) months	A nutrition support protocol addressing nutrition status and requirements, EN advancement and tolerance, and PN indication was implemented and evaluated for adherence and efficacy	After guideline implementation, a decrease in patients without nutrition support (3.1% vs 9.4%; $P = 0.043$), a delay in PN initiation (1 vs 3 days; $P = 0.027$), no difference in timing of EN initiation or achievement of energy and protein goals was noted	In the subgroup of protocol compliant (69.7%) vs noncompliant, the time to achieve goal energy (59 vs 103 h; $P < 0.001$) and target protein intake decreased (64 vs 135 h; $P = 0.02$) Outcomes limited by retrospective design
Miserachs et al ⁷⁴	Retrospective study; single center	To evaluate the impact of a standardized postoperative EN protocol on EN initiation and adequacy in children admitted to PICU immediately following liver transplant	n = 81 Median (IQR) age, 21.5 (9–72) months n = 49 preintervention n = 32 postintervention	Implementation of postoperative feeding protocol specifying timing and rate of EN initiation and advancement, formula type, and assessment of GI intolerance	Following protocol implementation, a decreased time to initiation (2 vs 3 days; $P = 0.005$) and achievement of goal EN via feeding tube ($P = 0.03$) and reduced PN use (47% vs 75%; $P = 0.01$)	No adverse events No differences in clinical outcomes Limited ability to analyze all pertinent data with retrospective design
Kyle et al ⁷⁵	Retrospective study; single center	To analyze energy and protein intake in infants and children admitted to PICU before and after implementation of nutrition support guidelines	n = 520 n = 335 preintervention n = 185 postintervention	Implementation of nutrition support guidelines including energy and protein requirements, EN algorithm, and nutrition support route of delivery	Protein intake was higher ($P \leq 0.05$) and energy intake higher in age >2 years ($P \leq 0.05$) following guideline implementation	Shorter time after introduction of the guideline and retrospective design limit ability to evaluate data impacting outcomes
Martinez et al ⁷⁰	Prospective, observational multicenter cohort study; 31 PICUs	To investigate the use and composition of EN algorithms and describe EN practices in infants and children requiring mechanical ventilation from 31 PICUs	n = 524 n = 431 received exclusive EN Mean (SD) age, 4.02 (4.91) years	Obtained details of any EN algorithms, EN energy and protein delivery, and determined adjuncts associated with improved EN adequacy	EN algorithms were used by 9 PICUs Use of antacids was associated with increased EN adequacy Use of EN algorithms, promotility agents, and postpyloric route was not associated with EN adequacy	Only 29% of PICUs used an EN guideline Enrollment restricted to children requiring mechanical ventilation and ≥ 8 -bed PICUs, potentially limiting wider applicability

TABLE 7 (Continued)

Reference	Study design; number of sites	Study aims	Patient characteristics (n)	Intervention	Outcomes	Comments
Wong et al ⁷¹	Systematic review of RCTs, before-and-after studies, prospective observations, and audits	To evaluate the impact of EN algorithms on clinical and nutrition outcomes in term infants and children admitted to the PICU >24 h	n = 9 studies n = 1564 children	Analyzed effect of EN algorithms on clinical and nutrition outcomes	Use of EN algorithm reduces GI intolerance and infections and improves timing of EN initiation and adequacy	The quality of the evidence assessed as low, owing to small numbers, heterogeneity, lack of control group, and concern for performance bias
Mehra et al ⁷⁶	Prospective, observational multicenter cohort study; 31 PICUs	To examine factors influencing the adequacy of energy and protein intake in the PICU and their relationship to clinical outcomes in infants and children requiring mechanical ventilation	n = 500 Mean (SD) age, 4.5 (5.1) years	Nutrition practices (including if a nutrition algorithm was used) were recorded for a maximum of 10 days Patients were followed for 60 days or until hospital discharge	A reduced rate of acquired infections was noted in patients in whom nutrition algorithms are used independent of protein and energy intake (OR, 0.18; 95% CI, 0.05–0.64; <i>P</i> = 0.008)	Only 29% of PICUs used an EN guideline
Meyer et al ⁷⁸	Time series study; single center	To examine the effect of implementing a series of updated EN protocols on nutrition practice in the PICU over 9 years	n = 353 over 4 audits in 9 years	Baseline assessment of EN delivery followed by serial implementation of EN protocol detailing EN route, initiation, advancement and monitoring, and audits of practice	Time to EN initiation decreased from median 15 to 4.5 h (<i>P</i> < 0.001) Use of PN decreased from 19% to 1% (<i>P</i> = 0.001)	Clinical outcomes not reported The decrease in PN use was observed during a portion of the audit years
Petrillo-Albarano et al ⁷²	Retrospective study; single center	To evaluate the effects of an EN protocol on tolerance and time to achieve target EN intake in the PICU	n = 184 n = 91 preintervention Mean (SD) age, 55.9 (70.9) months n = 93 postintervention Mean (SD) age, 68.1 (74.8) months	Implementation of an EN protocol specifying type of formula, EN initiation with advancement to goal rate, and standardized bowel regimen	After protocol introduction vs before guideline, median (IQR) time to EN initiation was shorter (14 h [9–21.5 h] vs 32 h [12–78 h]; <i>P</i> < 0.0001), and diarrhea (<i>P</i> = 0.009) and constipation (<i>P</i> = 0.012) improved	A short time period of 2 months after introduction of the guideline and retrospective design limits ability to evaluate data impacting outcomes

Abbreviations: CI, confidence interval; EN, enteral nutrition; GI, gastrointestinal; GRV, gastric residual volume; IQR, interquartile range; OR, odds ratio; PICU, pediatric intensive care unit; PN, parenteral nutrition; RCT, randomized controlled trial.

Grading of Recommendations Assessment, Development and Evaluation approach.⁷⁵ Low-quality evidence suggests that the use of a PICU EN algorithm is associated with improved gastrointestinal symptoms, time to initiation and adequacy of intake. No significant differences were seen in duration of mechanical ventilation, PICU or hospital length of stay, or PICU mortality, and hospital mortality was not reported by any of the studies with the use of an EN protocol vs the control group. Additional evidence suggests use of an EN protocol may impact infectious complications. A multicenter study of 500 children with critical illness from 31 PICUs demonstrated a significant association of reduced in-hospital-acquired infections in units using a nutrition support algorithm, independent of protein and energy intake.²⁶

EN algorithms have been reported in single-center studies to increase early and adequate EN in children with critical illness. Nutrition outcomes were evaluated in 91 children with critical illness before implementation of an EN algorithm, compared with 93 after the guideline was applied.⁷³ After use of the algorithm, the time to commencement of EN was significantly shorter with a median initiation of 14 h vs 32 h. Use of an EN protocol was investigated as part of a quality improvement initiative to increase EN intake after liver transplant in 49 children before and 32 children following introduction of the guideline.⁷⁶ Authors reported significantly decreased time to initiation and achievement of goal EN with a reduction in the use of PN. No adverse outcomes or differences were demonstrated in length of hospital stay or readmission rates. Following implementation of a PICU EN algorithm, cumulative 8-day energy and protein intakes were significantly improved in 185 children compared with 385 prior to the use of the guideline.⁷⁷ In another study of 162 children with critical illness, compliance with a nutrition support protocol following implementation significantly decreased the time to achieve target energy intake from 103 to 59 h and goal protein intake from 135 to 64 h.⁷⁸ Once a nutrition support guideline is implemented, routine measurement of performance is required to sustain improvements.⁶⁹ Meyer and colleagues performed four serial studies of 400 children requiring mechanical ventilation over a 9-year period, with each audit followed by introduction and modification of nutrition support protocols.⁷⁹ Improvements included significantly decreased time to initiate nutrition support from 15 h to 4.5 h, reduced use of PN from 11% to 4% of patients, and an increased proportion of patients receiving 50% and 70% energy goal by day 3. Although studies demonstrate improvements in EN delivery with the implementation of an EN guideline, research has not shown consistent improvements in clinical outcomes.

Nutrition support for infants with critical illness is typically initiated with a standard concentration of human milk or infant formula of 67 kcal/100 ml.⁷² Once target EN volume or fluid limit is met, energy density can be increased and protein supplements added to provide the target intake. With this strategy, infants required a significantly longer period of time to achieve goal protein and energy intake compared with children receiving a 100 kcal/100 ml pediatric formula.⁸⁰

Several studies have investigated the effects of protein- and energy-enriched vs standard EN for infants with critical illness through the use of high-calorie and high-protein infant formulas or human-milk fortification as summarized in Table 8. Infants with congenital heart disease following surgical repair receiving enriched EN revealed significantly improved adequacy of energy intake,⁸³ weight gain velocity,^{81–83} weight-for-age Z score,^{81,82,84} and nitrogen balance vs those provided standard formula.⁸⁵ A small study of 20 infants with critical illness randomly assigned to protein-energy-enriched formula demonstrated significantly increased energy and protein intakes, improved nitrogen balance, no adverse events, and no clinical outcome differences compared with those assigned to standard formula.⁸⁶ Some studies reported similar or increased gastrointestinal symptoms with protein-energy-enriched vs standard EN although none reached statistical significance. These small studies imply the possibility of positive nutrition effects but were not powered to detect differences in clinical outcomes.

Although many studies administer EN as a continuous modality, bolus feeding is an alternative that may affect outcomes. In a multicenter study of 147 children requiring mechanical ventilation in seven PICUs, patients were randomly assigned to receive continuous or bolus gastric EN.⁸⁷ The bolus modality resulted in a significantly shorter time to meet the EN goal with a subsequent increase in percentage of target energy and protein intake, without a difference in serial oxygen saturation index compared with the continuous group. A sample size of 430 for each group was desired to allow for a meaningful statistical analysis of outcomes; however, enrollment was slow and the study was terminated early at the request of the data safety monitoring board.

The optimal timing of supplemental or sole-source PN between days 2 and 7 of admission to the PICU is currently unknown. In a consensus article, ASPEN recommends consideration of PN initiation within 1–3 days in infants and 4–5 days in children and adolescents when EN is contraindicated or not feasible.⁸⁸ When exclusive EN is unable to meet $\geq 50\%$ of energy and protein targets, supplemental PN may be considered. A

TABLE 8 EN formulas.

Reference	Study design; number of sites	Study aims	Patient characteristics (n)	Intervention	Outcomes	Comments
<i>RCTs</i>						
Aryafar et al ⁸⁰	Open-label RCT; single center	To examine anthropometric and biochemical effects and GI tolerance of a high-energy-density EN formula compared with a standard-density formula administered by mouth to infants with CHD following corrective surgery	n = 64 n = 32 intervention group Median (IQR) age, 7 (6–8) months n = 32 control group Median (IQR) age, 7.5 (6.5–8) months	Randomized to receive 120 kcal/kg with high-energy-density formula (90 kcal/100 ml) vs standard-density formula (67 kcal/100 ml) from PICU discharge to 8 weeks after hospital discharge Phone contact 1 week after hospital discharge to review formula mixing, amounts administered, and GI tolerance	Infants receiving high-energy formula vs standard formula had improved weight gain ($P = 0.0001$), weight-for-age Z score ($P = 0.0001$), mid-upper arm circumference ($P = 0.0001$), less faltering of linear growth ($P = 0.0001$), increased serum albumin level ($P = 0.012$) No deaths or differences in energy or protein intake at week 1	2 patients in each group did not complete study protocol Parents of 4 infants receiving high-energy-density formula reported constipation Formula intake was only assessed at 1 week after hospital discharge
Scheeffer et al ⁷⁹	Open-label RCT; single center	To compare nutrition and clinical outcomes and tolerance of energy-enriched formula vs standard formula provided by mouth or via NGT to infants and children for 30 days following CHD surgery	n = 59 n = 29 protein-energy-enriched formula Mean (SD) age, 7.28 (6.4) months n = 30 standard formula Mean (SD) age, 6.25 (7.08) months	Randomized to receive energy- and protein-enriched formula (100 kcal/100 ml, 2.6 g protein/100 ml) vs standard energy and protein formula (67 kcal/100 ml, 1.4–2 g protein/100 kcal) from ICU discharge to 30 days after CHD surgery Target intake 120 kcal/kg or 150 kcal/kg for patients with low weight for length	Patients receiving enriched vs standard formula had increased weight-for-age Z scores ($P = 0.042$) and weight gain ($P = 0.03$), less frequent antibiotic use ($P = 0.047$), and higher incidence of diarrhea ($P = 0.01$) No difference in mortality or energy intakes	Diarrhea more frequent with higher osmolality, protein-energy-enriched formula Only 34% of protein-energy-enriched group and 50% of standard-formula group completed 30 days of study protocol because of death, clinical instability, cow milk protein allergy, chylous effusion, sepsis, GI intolerance, or nonacceptance

(Continues)

TABLE 8 (Continued)

Reference	Study design; number of sites	Study aims	Patient characteristics (n)	Intervention	Outcomes	Comments
Cui et al ⁸¹	Double-blind RCT; single center	To evaluate nutrition outcomes and tolerance of a protein- and energy-enriched formula vs standard formula administered to infants via continuous NGT for 5 days following CHD surgery	n = 50 n = 26 protein- and energy-enriched formula Mean (SD) age, 4.69 (3.54) months n = 24 standard formula Mean (SD) age, 5.23 (2.49) months	Randomized to receive 130 ml/kg with protein- and energy-enriched formula (100 kcal/100 ml, 2.6 g protein/100 ml) vs standard formula (67 kcal/100 ml, 1.4 g protein/100 ml) for 5 days in the CICU following CHD surgery	Infants receiving enriched formula vs standard formula had increased energy and protein intakes ($P < 0.05$; $P < 0.01$ for protein on day 1), higher nitrogen balance on days 2–5; ($P < 0.01$), and increased stool frequency on day 3 ($P = 0.03$) No deaths, no differences in duration of mechanical ventilation, CICU or hospital stay	2 patients did not complete the study because of CICU discharge 3 days after surgery Potential risk for overfeeding affecting outcomes
van Waardenburg et al ⁸²	Double-blind RCT; multicenter (2 PICUs)	To investigate nutrition outcomes, tolerance, and safety of protein- and energy-enriched formula vs standard formula via continuous NGT or nasoduodenal tube for 5 days in infants with RSV admitted to PICU requiring mechanical ventilation	n = 20 n = 8 protein- and energy-enriched formula Mean (SD) age, 2.7 (0.5) months n = 10 standard formula Mean (SD) age, 3.0 (0.6) months	Randomized to receive 130 ml/kg with protein- and energy-enriched formula (100 kcal/100 ml, 2.6 g protein/100 ml) vs standard formula (67 kcal/100 ml, 1.4 g protein/100 ml) for 5 days in the PICU	Infants receiving enriched formula vs standard formula had increased energy and protein intakes on days 3–5 ($P < 0.05$ for energy on days 3–4 and protein on day 3, all other $P < 0.01$), higher day 2–5 ($P < 0.05$) and cumulative nitrogen balance ($P < 0.001$), increased daily assessed by indirect calorimetry ($P < 0.01$), and higher GRVs ($P < 0.01$) No differences in gastric distention, diarrhea, emesis	2 infants from protein- and energy-enriched formula group excluded after study violation Nitrogen balance not completed on day 1 because of incomplete specimen collection Significant potential for overfeeding with positive energy balance as assessed by indirect calorimetry in 14 infants (5 in the enriched-formula group, 9 in the standard-formula group)

TABLE 8 (Continued)

Reference	Study design; number of sites	Study aims	Patient characteristics (n)	Intervention	Outcomes	Comments
Pillo-Blocka et al ⁷⁸	Double-blind RCT; single center	To evaluate weight gain and duration of hospital stay in infants following complete repair of CHD with intervention of rapid advancement of formula energy concentration vs control (usual practice) with standard advancement of formula energy concentration via NGT or oral intake until hospital discharge	n = 24 n = 16 intervention Mean (SD) age, 0.42 (0.27) months n = 18 control Mean (SD) age, 0.56 (0.25) months	Randomized after ICU discharge to rapid advancement with day 1 formula at 80 kcal/100 ml, day 2 at 90 kcal/100 ml, day 3 to hospital discharge at 100 kcal/100 ml, vs standard advancement with day 1 formula at 67 kcal/100 ml, day 3–5 at 80 kcal/100 ml, and day 6 to hospital discharge 90 kcal/100 ml	All received bolus EN via NGT at study entry and oral intake by hospital discharge Infants assigned to rapid vs standard advancement had higher energy adequacy (median, 98% vs 78% goal; $P = 0.01$) and weight gain ($P = 0.03$) over 3 days prior to hospital discharge, and shorter hospital stay ($P < 0.05$)	Only 70% of rapid-advancement group and 78% of standard-advancement group completed study protocol because of change in consent with a decision to breastfeed, unavailable study formula, and GI intolerance, without differences between groups

Observational studies

Eveleens et al ⁸³	Retrospective, observational cohort study; single center	To examine weight-for-age Z scores and GI symptoms in infants receiving a protein- and energy-enriched formula via postpyloric tube or oral intake for ≥ 2 weeks in the PICU	n = 70 Median (IQR) age, 76 (30–182) days	Demographic and clinical data collected for infants receiving protein- and energy-enriched formula (100 kcal/100 ml, 2.6 g protein/100 ml)	Mean (SD) weight-for-age Z score increased by 0.48 (1.10; $P < 0.001$) Enriched formula was stopped in 11% of patients for emesis, GRV > 50% of prior 4-h EN volume, and signs of discomfort	GRV may be an insensitive marker of GI intolerance Lack of control group receiving standard protein and energy formula and retrospective design limit clinical significance of outcomes
Babbitt et al ⁷⁷	Retrospective, observational; single center	To evaluate ability to attain goal energy intake and associated	n = 190 Mean (SD) age, 3.2 (5) years	Demographic and clinical data collected for children receiving	96% of patients attained target energy intake Time to reach goal energy was longer with age <1 year vs >1 year (41.7 ± 30 h vs 24.1 ± 34 h; $P = 0.008$)	Delay in achieving adequate energy intake limited by use of

(Continues)

TABLE 8 (Continued)

Reference	Study design; number of sites	Study aims	Patient characteristics (n)	Intervention	Outcomes	Comments
		complications in patients admitted to PICU and receiving EN via a transpyloric tube	n = 148 (78%) required mechanical ventilation	transpyloric EN in the PICU	Adverse events included 1 death due to intestinal perforation and 4 (2.1%) patients with necrotizing enterocolitis	lower-energy-density formula in infants vs children received higher-energy-density formula Lack of control group receiving EN via NGT Retrospective design limits ability to collect all data to evaluate outcomes

Abbreviations: CHD, congenital heart disease; CI, confidence interval; CICU, cardiac intensive care unit; EN, enteral nutrition; GI, gastrointestinal; GRV, gastric residual volume; ICU, intensive care unit; IQR, interquartile range; NGT, nasogastric tube; PICU, pediatric intensive care unit; PN, parenteral nutrition; RCT, randomized controlled trial; RSV, respiratory syncytial virus.

practical approach is to individualize supplemental PN to attain 60% of cumulative energy and protein goals by the end of the first week in the PICU.²⁸

LIMITATIONS AND RESEARCH GAPS

Evidence on the timing of EN initiation is limited by the small number of randomized controlled trials, mainly including patients with burn injuries, and thus results may not be applicable to all children with critical illness. Pediatric critical care research is characterized by small and inadequately powered trials because of limited patient volume and a reduced incidence of mortality compared with adults.⁸⁹ In addition, the heterogeneity of the PICU patient population in many studies may limit applicability to bedside practice.

The principal randomized controlled trial for timing of PN in the PICU, the PEPaNIC study, is remarkable in design and execution but has noteworthy limitations. Children with severe malnutrition and patients unable to receive EN were underrepresented in the study.⁴⁷ These populations may benefit from supplemental PN within 3–5 days of PICU admission.⁹⁰ PN initiation within 24 h of PICU admission is considered aggressive and not in accordance with accepted practice. As the trial compared PN on day 1 vs day 8, an opportunity has emerged to study initiation between days 2 and 7, particularly in more vulnerable infants and children.⁸⁹

More than 60% of patients in the PEPaNIC study receiving early PN were discharged from the PICU after day 3, which questions whether this short-term exposure to PN independently causes short- and long-term adverse effects.⁶⁰ Additionally, >75% of the late PN group was discharged before day 7 and never received PN.⁹¹ Many patients appeared to receive supplemental PN without adequate indication during a short PICU stay.⁴⁸ Goulet and colleagues further suggest overuse of PN, overfeeding, and lack of standardization of glucose control and nutrition delivery may have influenced the adverse effects of early PN.

Many research gaps exist in the timing of nutrition support in infants and children with critical illness. The current recommendation by SCCM-ASPEN is to meet at least two-thirds of cumulative EN goal by the end of the first week of PICU admission.¹ However, basic knowledge is lacking to accurately estimate energy needs for children with critical illness. Indirect calorimetry, the recommended standard to determine energy requirements, is not always feasible or available. As predictive equations are inaccurate in this population and can lead to overfeeding and underfeeding with resultant poor outcomes,^{1,89} a reliable estimate of energy needs is a

ground-level research gap impacting the timing and provision of nutrition support. Accurate reassessment during all phases of critical illness is also paramount to account for individual and dynamic changes over time.⁸⁹

The significance of timing of protein initiation and dose are high research priorities for children with critical illness.⁸⁹ Guidelines recommend a minimum of 1.5 g/kg protein,^{1,2} but the optimal strategy to achieve this intake is unknown. Evidence to guide appropriate carbohydrate and fat intakes in children during critical illness is also lacking. Enteral or parenteral route of administration, high vs low dose, and protein dose assessment methods all have opportunities to impact outcomes. Potential effects of combined early mobilization and early protein supplementation on clinical and nutrition outcomes are also relevant to the timing of nutrition support interventions in the PICU.

Timely provision of EN can be prevented by invasive and noninvasive positive pressure ventilation, hemodynamic instability, and postinjury or postoperative state.^{24,64} After initiation, EN is held because of intolerance and other clinical priorities for 30%–40% of the time in the PICU, negatively impacting time to achieve energy and protein goals.^{58,65} Research and consensus are needed to determine safety of EN during these conditions, as some obstacles are avoidable. The assessment of feeding intolerance is often subjective and requires better definition, consensus, and solutions.⁸⁹ Screening and early detection of gastrointestinal intolerance may allow for individualization of nutrition care to reduce the time to achieve EN intake goals. The potential role of probiotics, prokinetics, gastric vs postpyloric route, and formula composition are areas of interest to determine impact on gastrointestinal tolerance and, subsequently, the time to achieve EN targets. Knowledge regarding gastric motility during critical illness and techniques to accurately diagnose intolerance at the bedside are further areas in need of exploration.⁹²

A recent multicenter trial of children randomly assigned to bolus vs continuous gastric EN found a shorter time to reach goal intake and an increased target energy and protein intake in the bolus feeding group.⁸⁷ Further research to investigate EN modality and initiation of PN between days 2 and 7 is necessary to determine the best methods to optimize timing and achievement of nutrition support targets while minimizing the risk of harm.⁸⁹

CONCLUSION

Initiation of early EN within 24–48 h and achieving a target intake in the first week of critical illness are associated with improved survival, whereas early PN in

<24–48 h of PICU admission may be harmful to infants and children. Strategies to improve energy and protein intake require multidisciplinary efforts, and additional work is necessary to reveal outcomes related to the timing of nutrition support interventions in the PICU. Standardizing interventions and outcomes in future nutrition studies will improve validity and applicability of research to bedside practice.

AUTHOR CONTRIBUTIONS

Heather E. Skillman contributed to the conception and design of the review; Donna M. Fell, Emily A. Bitetto, and Heather E. Skillman equally contributed to the acquisition and analysis of the data, drafted and critically revised the manuscript, gave final approval of the manuscript, agree to be fully accountable for ensuring the integrity and accuracy of the work, and read and approved the final manuscript.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

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